# Euler discretizations, invariant faces and brackets for lower exponents of positive switched systems

Francesco Paolo Maiale
Gran Sasso Science Institute
Viale F. Crispi 7, 67100 L'Aquila, Italy
francescopaolo.maiale@gssi.it

**Abstract**

In this paper, we prove that explicit and implicit Euler discretizations yield monotone brackets for the lower Lyapunov exponent of positive switched systems. We establish convergence of the implicit bound for every Metzler family. For the explicit bound, we derive convergence criteria in reducible systems from a face formula for the uniform lower exponent on an invariant cone. The formula takes the minimum of the rates on faces that meet the cone nontrivially and are preserved by at least one mode, using only their preserving modes. For individually irreducible modes, we prove $O(h)$ discretization errors, where $h$ is the step size, and show that this order is sharp in general. At interior two-mode chattering optima with an irreducible optimal mixture, we obtain one-sided first-order bounds. We also construct a three-mode example in which the explicit bound converges while finite polytope lower bounds remain capped strictly below the true exponent. We prove convergence of fixed-antinorm Gripenberg brackets under individual irreducibility as the step and tolerance decrease and the search depth increases appropriately. Finally, we obtain independently verified continuous-time brackets by exact rational checks, with lower bounds that remain valid at smaller steps.

**Keywords:** Positive switched systems, Lower Lyapunov exponent, Lower spectral radius, Euler discretization, Invariant faces, Polytope antinorms.

## 1 Introduction

Let $\mathcal{A} = \{A_1, \dots, A_m\}$ be a family of Metzler matrices. Then the lower exponent $\check{\lambda}(\mathcal{A})$ is the best exponential growth rate achievable by switching in $\dot{x} = A_{\sigma(t)}x$ from interior initial states (Proposition 2.2). The computation of rigorous bounds requires estimating discretization error and verifying the discrete bounds. For reducible systems, boundary dynamics can limit the rates established by uniform lower-growth functions even when the discretization converges to $\check{\lambda}$.

The discrete lower spectral radius admits polytope bounds [12] and Gripenberg-type bounds [9]. For continuous-time systems, Guglielmi, Laglia and Protasov [12, Sec. 4] obtain a $C\tau + \varepsilon$ gap using $\mathrm{e}^{\tau A}$ and suitable polytope antinorms for individually $K$-irreducible generators, where $\varepsilon$ is the polytope approximation error.

Earlier methods use irreducibility or secondary-eigenvector conditions [12, 9]; boundary eigenvectors can obstruct polytope methods [10, Sec. 6.4].

### 1.1 Main results and conventions

We first summarize the main results obtained in this paper, first theoretical and then numerical ones.

(R1) *Face formula* (Theorem 2.7). The uniform lower exponent on an invariant cone equals the minimum of the exponents of the faces it touches that are preserved by at least one mode, each restricted to its preserving modes. This identifies the boundary rates above. Occupation measures [23, 7, 8] make the face decomposition explicit within the ergodic variational framework [4, 5, 13]. Reduced graph classes give the one-matrix analogue [6, Thms. 5.10 and 6.8].

(R2) *Monotone Euler brackets* (Theorems 3.1 and 3.4). Explicit and implicit Euler steps give $g(h) \le \check{\lambda} \le G(h)$ that improve monotonically as $h \downarrow 0$. The implicit bound always converges. The explicit limit lies between the boundary exponent and $\check{\lambda}$; cone criteria based on the face formula give convergence in reducible systems, verified on five test pairs (Corollary 3.8 and Computation 3.9). Compared with the exponential discretization of [12, Sec. 4], Euler steps add monotonicity in $h$, convergence of the implicit bound without irreducibility, and polytope lower bounds that remain valid at smaller steps. The explicit bound parallels classical stability under smaller Euler steps [17, 16, 18].

(R3) *Rates* (Theorems 3.5 and 3.12). For individually irreducible modes both errors are $O(h)$, with this order optimal in general. At interior two-mode chattering optima with an irreducible optimal mixture we obtain one-sided first-order coefficients. Related work treats polyhedral Lyapunov functions shared by linear systems and their Padé discretizations [19], and second-order bounds for the upper Lyapunov exponent under positive dwell-time constraints [15].

(R4) *Limits of polytope bounds* (Lemma 2.3 and Example 4.6). Lower-growth functions comparable to $\mathbf{1}^\top x$ cannot establish rates above the boundary exponent. In a three-mode example, we show that the explicit Euler bound converges while lower bounds obtained from finite polytopes remain capped strictly below $\check{\lambda}$.

(R5) *Rigorous brackets* (Propositions 4.3 and 4.4 and Corollary 4.1). Building on polytope antinorm methods [10, 12, 11, 9], rational polytopes, vector chains and exponential enclosures yield continuous-time brackets verified independently, also for reducible families; exact verification also appears for the joint spectral radius [14]. Fixed-antinorm Gripenberg brackets converge for individually irreducible modes as $h \downarrow 0$, with tolerance $\delta = o(h)$ and sufficient search depth.

The convergence results are summarized below; $\check{\lambda}_{\min} \le \check{\lambda}$ denotes the boundary exponent (Section 2.2).

| | general Metzler family | individually irreducible modes |
|---|---|---|
| expl. $g(h)$ | $g(h) \le \check{\lambda}$, monotone; $\check{\lambda}_{\min} \le \lim g \le \check{\lambda}$; $\lim g = \check{\lambda}$ under the cone criteria; open in general (Theorem 3.1 and Corollary 3.8) | $\check{\lambda} - Ch \le g(h)$ (Theorem 3.5); order $h$ sharp (Example 3.10) |
| impl. $G(h)$ | $\check{\lambda} \le G(h)$, monotone, $G(h) \to \check{\lambda}$; no rate proved (Theorem 3.4) | $G(h) \le \check{\lambda} + Ch$ (Theorem 3.5); order $h$ sharp |

Experiments up to dimension 20 distinguish discretization and search effects and compare with Algorithm (L) of [12]. Numerical schedule rates can fall below numerically minimized constant-mixture rates. Note, however, that these comparisons do not prove strict gaps from all constant mixtures.

**Organization of the paper** Section 2 develops the face formula, Section 3 proves the Euler bounds, Section 4 treats rigorous bounds and numerics, and Section 5 discusses some open problems. The appendices give the test matrices (Appendix A), Gripenberg analysis and

detailed runs (Appendix B), first-order diagnostics (Appendix C), and the two technical proofs (Appendices D and E). Experimental protocols are collected in Appendix F, and the larger numerical samples in Appendix G.

## 2 Lower exponents and invariant faces

In this section, we define the lower exponent, construct the lower-growth functions used in Section 3, and prove our main result (the face formula).

### 2.1 Switched systems and lower exponents

Let $\mathcal{A} = \{A_1, \dots, A_m\} \subset \mathbb{R}^{d\times d}$ be a family of Metzler matrices, i.e. with nonnegative off-diagonal entries, and let $K = \mathbb{R}^d_+$ be the non-negative cone. We denote the simplex associated to $K$ by $\Delta = \{x \in K \ : \ \mathbf{1}^\top x = 1\}$.

**Definition 2.1.** A *switching signal* $\sigma$ is any measurable function $[0,\infty) \to \{1,\dots,m\}$.

For $t \geq s \geq 0$, let $\Phi_\sigma(t,s)$ be the transition matrix of $\dot{x} = A_{\sigma(t)}x$ from time $s$ to time $t$, and write $\Phi_\sigma(t) = \Phi_\sigma(t,0)$. Notice that every $\Phi_\sigma(t,s)$ is nonnegative. Set

$$\bar{\mu}(t) = \inf_\sigma \|\Phi_\sigma(t)\|_1, \qquad \check{\lambda}(\mathcal{A}) = \lim_{t\to\infty} \frac{\log \bar{\mu}(t)}{t}.$$

For a Metzler matrix $S$, let $\eta(S)$ denote its spectral abscissa. Let $S_\alpha = \sum_i \alpha_i A_i$ for $\alpha$ in the control simplex $\Delta_m = \{u \in \mathbb{R}^m_+ \ : \ \sum_i u_i = 1\}$. Throughout, $m$ is the number of modes, and

$$K_A = \max_{1\leq i\leq m} \|A_i\|_1, \qquad c_A = \max_{\substack{1\leq i\leq m\\ 1\leq j\leq d}} |(A_i)_{jj}|.$$

**Proposition 2.2.** *The following properties hold:*

(i) *The limit $\check{\lambda}$ exists, equals $\inf_{t>0} t^{-1}\log\bar{\mu}(t)$ and does not depend on the norm. In addition, $|\check{\lambda}| \leq K_A$.*

(ii) *For every $x_0 > 0$, $\check{\lambda} = \lim_t t^{-1}\log\inf_\sigma \|\Phi_\sigma(t)x_0\|_1$. Hence $\check{\lambda}$ is the best exponential rate achievable from any interior initial state, by open-loop or by state-feedback switching.*

(iii) *For every $\alpha \in \Delta_m$, $\check{\lambda} \leq \eta(S_\alpha)$. For every rational $x > 0$,*

$$\eta(S_\alpha) \leq \max_j \frac{(S_\alpha x)_j}{x_j} \qquad \text{(Collatz–Wielandt inequality).}$$

*This upper bound is rational if the matrices $A_i$ and the chosen weights $\alpha_i$ are rational.*

*Proof.* Concatenating switching signals shows $\bar{\mu}(t+s) \leq \bar{\mu}(t)\bar{\mu}(s)$. Since $\|\Phi_\sigma(t)^{\pm 1}\|_1 \leq e^{K_A t}$, we have $e^{-K_A t} \leq \bar{\mu}(t) \leq e^{K_A t}$. Fekete's lemma gives the limit and the infimum formula, while equivalence of norms gives norm independence, proving (i).

For (ii), notice that given $\Phi \geq 0$ and $x_0 > 0$, it holds

$$\min_j (x_0)_j \, \|\Phi\|_1 \leq \mathbf{1}^\top \Phi x_0 \leq d \max_j (x_0)_j \, \|\Phi\|_1.$$

A feedback law produces, for a fixed initial state, an open-loop signal. For (iii), the Lie–Trotter formula (see [22]) gives

$$\left(\prod_i e^{(t/n)\alpha_i A_i}\right)^n \xrightarrow{n\to\infty} e^{tS_\alpha},$$

and each left-hand side is a transition matrix of a switching signal. Hence $\bar{\mu}(t) \leq \|e^{tS_\alpha}\|_1$, and $t^{-1}\log\|e^{tS_\alpha}\|_1 \to \eta(S_\alpha)$. The Collatz–Wielandt bound follows from the Perron–Frobenius theorem applied to $S_\alpha + c_A I \geq 0$ in the weighted norm $\max_j |y_j|/x_j$. □

## 2.2 Uniform exponents on cones

The exponent $\check{\lambda}$ describes optimal growth from each fixed interior state. To describe growth uniformly over normalized states in a cone, let $C \subseteq K$ be a closed convex cone, $C \neq \{0\}$, that every $e^{tA_i}$ maps into itself, and define

$$\mu_C(t) = \inf_{\sigma} \min_{x \in C \cap \Delta} \mathbf{1}^\top \Phi_\sigma(t)x, \qquad \check{\lambda}_C = \lim_{t\to\infty} \frac{\log \mu_C(t)}{t},$$

where the infima range over both signals and initial states in $C \cap \Delta$. Every $\Phi_\sigma(t)$ maps $C$ into $C$: for piecewise-constant $\sigma$ directly, and for measurable $\sigma$ by approximation, since $C$ is closed and $\Phi_\sigma(t)$ depends continuously on $\sigma$ in $L^1$. So, for $x \in C$, we have

$$\mathbf{1}^\top \Phi_\sigma(t+s)x \geq \mu_C(t)\, \mathbf{1}^\top \Phi_\sigma(s)x.$$

Hence $\mu_C$ is supermultiplicative, the limit exists and equals $\sup_t t^{-1} \log \mu_C(t)$. We thus write $\check{\lambda}_{\min} = \check{\lambda}_K$ and call it the boundary exponent since it controls the decay of every initial state, including those on the boundary of $K$.

**Lemma 2.3.** *The following properties hold:*

(i) $\check{\lambda}_{\min} \leq \check{\lambda}_C \leq \check{\lambda}$. *Moreover* $\check{\lambda}_{\min} \leq \eta(A_i[J,J])$ *whenever* $J \neq \emptyset$ *and* $A_i$ *leaves the face* $K \cap \operatorname{span}\{e_j : j \in J\}$ *invariant.*

(ii) *Let* $a : C \to [0,\infty)$ *satisfy* $c_1 \mathbf{1}^\top x \leq a(x) \leq c_2 \mathbf{1}^\top x$ *on* $C$ *for constants* $0 < c_1 \leq c_2$, *and* $a(\Phi_\sigma(t)x) \geq e^{\nu t} a(x)$ *for all* $\sigma$, $t \geq 0$ *and* $x \in C$. *Then* $\nu \leq \check{\lambda}_C$.

*Proof.* The first inequality holds because $C \cap \Delta \subseteq \Delta$. For $x \in C \cap \Delta$, $\mathbf{1}^\top \Phi_\sigma(t)x \leq \|\Phi_\sigma(t)\|_1$, so $\mu_C(t) \leq \bar{\mu}(t)$ and $\check{\lambda}_C \leq \check{\lambda}$. If the face is invariant, a nonnegative Perron vector $x_J$ of $A_i[J,J]$, extended by zero and normalized, satisfies $A_i x = \eta(A_i[J,J])\, x$, so

$$\mu_K(t) \leq \mathbf{1}^\top e^{tA_i} x = e^{\eta(A_i[J,J])t}.$$

For (ii), any $x \in C \cap \Delta$ satisfies $\mathbf{1}^\top \Phi_\sigma(t)x \geq a(\Phi_\sigma(t)x)/c_2 \geq e^{\nu t} a(x)/c_2 \geq (c_1/c_2)\, e^{\nu t}$. □

## 2.3 Positivity and lower-growth functions

Individual irreducibility removes the boundary obstruction by making transition matrices uniformly positive over each fixed positive time interval.

**Lemma 2.4.** *If every* $A_i$ *is irreducible, then for every* $\tau > 0$ *there is* $\delta > 0$ *with* $\Phi_\sigma(\tau) \geq \delta\, \mathbf{1}\mathbf{1}^\top$ *for every switching signal* $\sigma$. *Consequently,*

$$\check{\lambda}_{\min} = \check{\lambda}.$$

*Proof.* We may assume $d \geq 2$. Write $A_i = N_i - c_A I$, so that $N_i \geq 0$, and $\Phi_\sigma(t) = e^{-c_A t} \Psi_\sigma(t)$ where $\dot{\Psi} = N_\sigma \Psi$. The Dyson series has nonnegative terms, so $\Psi_\sigma(b,a) \geq I + \int_a^b N_\sigma$. The proof now follows from this sequence of steps:

- Split $[0,\tau]$ into $d-1$ intervals of length $\tau/(d-1)$. On each, some mode $i_k$ is active on a set of measure at least a fraction $1/m$ of the interval, so $\Psi_\sigma(\tau) \geq \prod_k (I + \kappa N_{i_k})$ with $\kappa = \tau/((d-1)m)$.
- Each $N_{i_k}$ is irreducible, so at every factor the support of each column of the product gains at least one index. After $d-1$ factors the product is positive.
- Every positive entry is a sum of products of at most $d-1$ entries of the $\kappa N_i$ and of $I$. So it is at least $(\min\{1,\kappa\}\nu_0)^{d-1}$, where $\nu_0 = \min\{1, \text{smallest positive entry of the } N_i\}$.

So the lemma holds with $\delta := e^{-c_A\tau}(\min\{1,\kappa\}\nu_0)^{d-1}$. Conversely, if some $A_i$ is reducible, then $e^{\tau A_i}$ has a zero entry, and no such $\delta$ exists. To conclude, for $x \in \Delta$ and $t \geq \tau$,

$$\mathbf{1}^\top \Phi_\sigma(t)x \geq \delta\, \mathbf{1}^\top \Phi_\sigma(t,\tau)\mathbf{1} \geq \delta\, \bar{\mu}(t-\tau),$$

so $\check{\lambda}_{\min} \geq \check{\lambda}$, and Lemma 2.3(i) gives the reverse inequality. □

The next lemma supplies the lower-growth functions used in the Euler estimates. Part (i) uses a finite time horizon for any $\nu < \check{\lambda}_C$. Under irreducibility, part (ii) attains the exact rate $\check{\lambda}$ and provides the Lipschitz control needed for the $O(h)$ error bound.

**Lemma 2.5.**

*(i) For every $\nu < \check{\lambda}_C$ there is a concave, positively homogeneous function $a_T$ on $\mathbb{R}^d$, Lipschitz with some constant $L$ in the $\ell^1$ norm, such that $0 \leq a_T(x) \leq C'\mathbf{1}^\top x$ on $K$, $a_T \geq c_0 > 0$ on $C \cap \Delta$, and $a_T(e^{sA_i}y) \geq e^{\nu s}a_T(y)$ for $y \in C \cap \Delta$ and $s \in [0,1]$.*

*(ii) If every $A_i$ is irreducible, then for every $\tau > 0$ the function $a(x) = \inf\{e^{-\check{\lambda}t}\mathbf{1}^\top\Phi_\sigma(t)x : t \geq \tau,\ \sigma\}$ satisfies $a(e^{sA_i}x) \geq e^{\check{\lambda}s}a(x)$ for all $x \in K$ and $s \geq 0$, and $c_a\mathbf{1}^\top x \leq a(x) \leq C_a\mathbf{1}^\top x$ on $K$ for some constants $0 < c_a \leq C_a$. Moreover $a$ is finite and Lipschitz on an $\ell^1$-neighbourhood $N'$ of $\Delta$, while outside $K \cup N'$ it may take the value $-\infty$.*

*Proof.* Set $\xi = (\check{\lambda}_C - \nu)/2$. Since $\Phi_\sigma(t) \geq e^{-c_A t}I$, we have $\mu_C(t) \geq e^{-c_A t}$. By the convergence $t^{-1}\log\mu_C(t) \to \check{\lambda}_C$, there is $c_0 > 0$ such that

$$e^{-\nu t}\mathbf{1}^\top\Phi_\sigma(t)y \geq c_0 e^{\xi t} \qquad \text{for all } y \in C \cap \Delta, \text{ all } \sigma \text{ and all } t \geq 0.$$

Fix $\tau > 0$, let $C_0 = \max_{y\in\Delta} e^{-\nu\tau}\mathbf{1}^\top e^{\tau A_1}y$, and choose $T \geq \tau$ with $c_0e^{\xi T} \geq C_0 + 1$. Define

$$a_T(x) = \inf\big\{e^{-\nu t}\mathbf{1}^\top\Phi_\sigma(t)x \ :\ \tau \leq t \leq T,\ \sigma\big\}, \qquad x \in \mathbb{R}^d.$$

It is an infimum of linear functionals whose coefficient vectors $e^{-\nu t}\Phi_\sigma(t)^\top\mathbf{1}$ are nonnegative and bounded by some $C'$. So $a_T$ is finite, concave, positively homogeneous and Lipschitz on $\mathbb{R}^d$, $0 \leq a_T(x) \leq C'\mathbf{1}^\top x$ on $K$, and $c_0 \leq a_T \leq C_0$ on $C \cap \Delta$. Concatenating signals gives

$$a_T(e^{sA_i}x) \geq e^{\nu s}\min\{a_T(x),\, b_s(x)\}, \quad b_s(x) = \inf\{e^{-\nu t}\mathbf{1}^\top\Phi_\sigma(t)x \ :\ T < t \leq T+s,\ \sigma\}$$

for $s \in (0,1]$ and every $x$. On $C \cap \Delta$ we have $b_s \geq c_0e^{\xi T} \geq a_T + 1$, so $a_T(e^{sA_i}y) \geq e^{\nu s}a_T(y)$ there, proving (i).

By Lemma 2.4 there is $\delta > 0$ with $\Phi_\sigma(\tau) \geq \delta\,\mathbf{1}\mathbf{1}^\top$ for every $\sigma$. Each coefficient vector $w = e^{-\check{\lambda}t}\Phi_\sigma(t)^\top\mathbf{1}$, $t \geq \tau$, has the form $w = \Phi_\sigma(\tau)^\top u$ with $u \geq 0$. So $\delta\|u\|_1 \leq w_j \leq M\|u\|_1$, where $M$ bounds $\Phi_\sigma(\tau)$ entrywise, and $\max_j w_j \leq \kappa_0 \min_j w_j$ with $\kappa_0 = M/\delta$. Then:

- *Lower bound.* By Proposition 2.2(i), $\bar{\mu}(t) \geq e^{\check{\lambda}t}$ so for $y \in \Delta$ we have

$$w \cdot y \geq \min_j w_j \geq (d\kappa_0)^{-1}\mathbf{1}^\top w = (d\kappa_0)^{-1}e^{-\check{\lambda}t}\mathbf{1}^\top\Phi_\sigma(t)\mathbf{1} \geq (d\kappa_0)^{-1}e^{-\check{\lambda}t}\bar{\mu}(t) \geq (d\kappa_0)^{-1}.$$

  So $c_a = (d\kappa_0)^{-1}$ works, while for $C_a$ one can choose $t = \tau$ and $\sigma \equiv 1$.

- *Lipschitz continuity.* Let $C_0'$ bound the functional with $t = \tau$ and $\sigma \equiv 1$ on

$$N' = \big\{x \ :\ \|x - y\|_1 \leq 1/(2\kappa_0) \text{ for some } y \in \Delta\big\}.$$

  For each $x \in N'$, choose $y \in \Delta$ with $\|x - y\|_1 \leq 1/(2\kappa_0)$. Then

$$w \cdot x \geq \min_j w_j - \|w\|_\infty\|x - y\|_1 \geq \tfrac{1}{2}\min_j w_j.$$

  So the near-minimizers, with $w \cdot x \leq a(x) + 1$, satisfy $\max_j w_j \leq 2\kappa_0(C_0' + 1) =: B$. Hence on $N'$, $a$ coincides with the infimum over coefficient vectors with $\max_j w_j \leq B$, which is Lipschitz with constant $B$.

- *Invariance.* Concatenation gives $a(e^{sA_i}x) \geq e^{\check{\lambda}s}a(x)$ for all $s \geq 0$ and $x \in K$. □

## 2.4 Invariant faces and the face formula

We now identify the boundary rates through the invariant faces and their preserving modes. For nonempty $J \subseteq \{1, \dots, d\}$ let $F_J = \{x \in K \ : \ x_k = 0 \text{ for } k \notin J\}$, and let

$$S(J) = \{i \ : \ e^{tA_i} F_J \subseteq F_J \text{ for all } t \geq 0\} = \{i \ : \ (A_i)_{kj} = 0 \text{ for all } k \notin J,\ j \in J\}$$

be the set of modes that leave $F_J$ invariant. If $S(J) \neq \emptyset$, let $\check{\lambda}_J$ be the lower exponent of the restricted family $\{A_i[J,J] \ : \ i \in S(J)\}$, on $\mathbb{R}^J$. Thus $\check{\lambda}_{\{1,\dots,d\}} = \check{\lambda}$ and

$$\begin{aligned} S(J) = \{i\} &\implies \check{\lambda}_J = \eta(A_i[J,J]) \\ J = \{j\} &\implies \check{\lambda}_J = \min_{i \in S(J)} (A_i)_{jj}. \end{aligned}$$

For a cone $C$ let $\Sigma(C) = \{\operatorname{supp} x \ : \ x \in C \setminus \{0\}\}$ be the set of supports of its points.

**Definition 2.6.** We call a face $F_J$ *fast* if $S(J) \neq \emptyset$ and $\check{\lambda}_J < \check{\lambda}$, and *slow* otherwise.

**Theorem 2.7** (Face formula)**.** *Let $C \subseteq K$ be a closed convex cone, $C \neq \{0\}$, invariant under every matrix exponential $e^{tA_i}$. Then*

$$\check{\lambda}_C = \min\big\{\check{\lambda}_J \ : \ J \in \Sigma(C),\ S(J) \neq \emptyset\big\},$$

*and the set on the right is nonempty. In particular, for $C = K$,*

$$\check{\lambda}_{\min} = \min\big\{\check{\lambda}_J \ : \ J \neq \emptyset,\ S(J) \neq \emptyset\big\},$$

*which corresponds to the minimum of the lower exponents of the faces of $K$ that are invariant under at least one mode, $K$ itself included.*

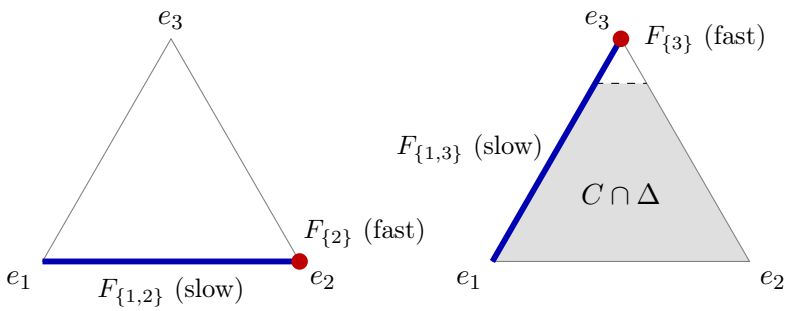
This formula involves the invariant faces that $C$ touches. Indeed, the minimum is unchanged if $\Sigma(C)$ is replaced by the set of all $J$ with $C \cap F_J \neq \{0\}$. If $y \in C \cap F_J \cap \Delta$ and $x \in F_J$ has support $J$, then $x \geq cy$ for some $c > 0$, so the exponent $\check{\lambda}_J$ of $x$ under signals in $S(J)$ is at least $\check{\lambda}_C$. For a single mode, this reduces to the classical description of the growth of $e^{tA}x$, $x \geq 0$, through the classes of the reduced graph [6].

As a consequence, by Lemma 2.3(ii), lower-growth functions comparable to $\mathbf{1}^\top x$ on $K$, including piecewise-linear ones, cannot establish rates above $\check{\lambda}_{\min}$.

Equivalently, $\check{\lambda}_C$ is the minimal exponent of $\dot{x} = (\sum_i u_i A_i)x$, $u(t) \in \Delta_m$, over initial states in $C$, attained by an ergodic control-flow measure [4, 5, 13]. An associated extremal trajectory has nondecreasing support, eventually some $J$. Then, its subsequent control uses only $S(J)$, giving $\check{\lambda}_C \geq \check{\lambda}_J$.

**Example 2.8.** Case 148 of Appendix A has two reducible modes, Figure 1 shows its invariant faces. As given, $A_1$ leaves $F_{\{1,2\}}$ invariant, with exponent $\eta(A_1[\{1,2\},\{1,2\}]) \approx -1.220815$, and $A_2$ leaves $F_{\{2\}}$ invariant, with exponent $(A_2)_{22} \approx -1.368010$; no proper face is invariant under both modes. By Theorem 2.7, $\check{\lambda}_{\min} = (A_2)_{22}$. Since $-1.247834 \leq \check{\lambda} \leq -1.241336$ (Remark 4.5), $F_{\{2\}}$ is fast and $F_{\{1,2\}}$ is slow. After transposition, which changes neither $\check{\lambda}$ nor the Euler bounds (Proposition 3.7(iv)), $A_1^\top$ leaves $F_{\{3\}}$ invariant, with exponent $(A_1)_{33} \approx -3.669288$ (fast), and $A_2^\top$ leaves $F_{\{1,3\}}$ invariant, with exponent $\approx 0.147543$ (slow). The case recurs three times: a cone touching only slow faces proves explicit Euler convergence (Computation 3.9); a polytope avoiding the fast face certifies a lower bound above $\check{\lambda}_{\min}$ (Remark 4.5); and the fast face caps every positive antinorm used by the Gripenberg algorithm (Appendix B).

The proof makes this decomposition explicit through projective occupation measures, as in long-run average control [23, 7, 8]. The key step, Lemma 2.9, restricts such measures to the points of support $J$, where only modes in $S(J)$ occur and mean growth is at least $\check{\lambda}_J$.




(a) as given (b) transposed

Figure 1: Invariant faces of case 148 (Example 2.8) on the simplex $\Delta$: slow faces as thick edges, fast faces as filled vertices. The shaded set $C \cap \Delta$ for the cone of Computation 3.9 touches the slow edge but not the fast vertex; it is drawn with $\beta = 1/5$ instead of $1/20$ for visibility.

### 2.4.1 Projective dynamics and occupation measures

Throughout the rest of this subsection, fix $C$ as in Theorem 2.7 and $X = C \cap \Delta$. For $x \in K \setminus \{0\}$ let $p = x/\mathbf{1}^\top x \in \Delta$. Under mode $i$, $p$ satisfies

$$\dot{p} = F_i(p) := A_i p - (\mathbf{1}^\top A_i p)\, p, \qquad \tfrac{\mathrm{d}}{\mathrm{d}t} \log \mathbf{1}^\top x = \gamma_i(p) := \mathbf{1}^\top A_i p,$$

so that, for $x \in \Delta$ and the projective solution $p(s)$, it holds

$$\mathbf{1}^\top \Phi_\sigma(t) x = \exp \int_0^t \gamma_{\sigma(s)}(p(s))\, \mathrm{d}s, \qquad |\gamma_i| \le K_A \ \text{ on } \Delta. \tag{1}$$

The set $X$ is compact and forward invariant under the projective dynamics. For $x \in X$, a switching signal $\sigma$ and $T > 0$, the measure $\frac{1}{T}\int_0^T \delta_{(p(s),\sigma(s))}\, \mathrm{d}s$ is a probability measure on $X \times \{1, \dots, m\}$. Let $\mathcal{M}$ be the set of weak$^*$ limits of such measures along sequences $(x_n, \sigma_n, T_n)$ with $T_n \to \infty$. Since the probability measures on the compact set $X \times \{1, \dots, m\}$ are weak$^*$ compact, every such sequence has a convergent subsequence. Since $|f(p(T)) - f(p(0))|/T \le 2\|f\|_{\infty,X}/T \to 0$ and $(p, i) \mapsto \nabla f(p) \cdot F_i(p)$ is continuous, every $\mu \in \mathcal{M}$ satisfies

$$\int \nabla f(p) \cdot F_i(p)\, \mathrm{d}\mu(p, i) = 0 \qquad \text{for every } f \in C^1(\mathbb{R}^d). \tag{2}$$

For $J \neq \emptyset$ let $X_J := \{p \in \Delta \ : \ \operatorname{supp} p = J\}$, and write $\mu(X_J)$ for $\mu(X_J \times \{1, \dots, m\})$. The sets $X_J$ partition $\Delta$, and $X \subseteq \bigcup_{J \in \Sigma(C)} X_J$.

**Lemma 2.9.** *Let $\mu \in \mathcal{M}$ and let $J \subseteq \{1, \dots, d\}$ be nonempty. Then:*

(a) *for $\mu$-a.e. $(p, i)$ with $p \in X_J$ we have $i \in S(J)$; in particular $\mu(X_J) = 0$ if $S(J) = \emptyset$;*

(b) $\int_{X_J} \nabla f \cdot F\, \mathrm{d}\mu = 0$ *for every $f \in C^1(\mathbb{R}^d)$;*

(c) *if $S(J) \neq \emptyset$, then $\int_{X_J} \gamma\, \mathrm{d}\mu \ge \check{\lambda}_J\, \mu(X_J)$.*

Here $\int_{X_J} \nabla f \cdot F\, \mathrm{d}\mu$ denotes the integral $\int_{X_J \times \{1,\dots,m\}} \nabla f(p) \cdot F_i(p)\, \mathrm{d}\mu(p, i)$, and similarly for $\gamma$. The proof is technical and thus given in Appendix E.

### 2.4.2 Proof of the face formula

*Proof of Theorem 2.7.* For simplicity, we divide the proof into upper and lower bound respectively.

**Upper bound.** Let $J \in \Sigma(C)$ with $S(J) \neq \emptyset$, and choose $x \in X$ with support $J$. Signals with values in $S(J)$ keep $\Phi_\sigma(t)x$ in $F_J$, so $\mathbf{1}^\top \Phi_\sigma(t)x \le \|\Phi_\sigma(t)|_{F_J}\|_1$. Taking their infimum gives $\mu_C(t) \le \bar{\mu}_J(t)$, where $\bar{\mu}_J(t) = \inf_{\sigma:\,\sigma(s)\in S(J)} \|\Phi_\sigma(t)|_{F_J}\|_1$, and hence $\check{\lambda}_C \le \check{\lambda}_J$.

We claim that the set in the minimum is nonempty. Fix a mode $i$. Its projective flow $P_i^t(v) = e^{tA_i}v/\mathbf{1}^\top e^{tA_i}v$ maps the compact convex set $X$ into itself. By Brouwer's theorem, the sets $E_n = \{v \in X :\ P_i^{2^{-n}}(v) = v\}$, $n \ge 0$, are nonempty and compact. The semigroup property gives $E_{n+1} \subseteq E_n$, so their intersection contains a point $v$. It is fixed at every nonnegative dyadic time, and hence at every $t \ge 0$ by continuity. Differentiation at $t = 0$ gives $A_i v = \beta v$ for $\beta = \mathbf{1}^\top A_i v$. If $J = \operatorname{supp} v$ and $k \notin J$, then $0 = (A_i v)_k = \sum_{j\in J}(A_i)_{kj}v_j$. Finally, nonnegativity and $v_j > 0$ imply $i \in S(J)$.

**Lower bound.** Let $\lambda_*$ be the minimum in the statement. For $\mu \in \mathcal{M}$, Lemma 2.9(a),(c) and $X \subseteq \bigcup_{J\in\Sigma(C)} X_J$ give

$$\int \gamma \,\mathrm{d}\mu = \sum_{J\in\Sigma(C),\ S(J)\neq\emptyset} \int_{X_J} \gamma \,\mathrm{d}\mu \ge \sum_J \check{\lambda}_J\, \mu(X_J) \ge \lambda_*.$$

Now let $\nu < \lambda_*$. If for every $n$ there were $x_n \in X$ and $\sigma_n$ with $\int_0^n \gamma_{\sigma_n}(p_n(s))\,\mathrm{d}s < \nu n$, a weak$^*$ limit point $\mu \in \mathcal{M}$ of their occupation measures on $[0, n]$ would satisfy $\int \gamma \,\mathrm{d}\mu \le \nu < \lambda_*$, a contradiction. So there is $T_0$ with $\int_0^{T_0} \gamma_\sigma(p(s))\,\mathrm{d}s \ge \nu T_0$ for all $x \in X$ and all $\sigma$. Since $X$ is forward invariant, (1) applied to consecutive blocks of length $T_0$ gives

$$\mathbf{1}^\top \Phi_\sigma(t)x \ge e^{\nu t - (|\nu| + K_A)T_0} \quad \text{for } x \in X \implies \check{\lambda}_C \ge \nu.$$

Letting $\nu \uparrow \lambda_*$ gives $\check{\lambda}_C \ge \lambda_*$. □

# 3 Discretization bounds

Explicit and implicit Euler steps give lower and upper bounds for $\check{\lambda}$. We first prove general bounds and implicit convergence, then an $O(h)$ bracket for irreducible modes and cone criteria for reducible families. Sharpness examples and one-sided first-order coefficients complete the section.

## 3.1 General Euler bounds

For a finite family $\mathcal{F} = \{M_1, \dots, M_r\}$ of square matrices, its lower spectral radius is

$$\check{\rho}(\mathcal{F}) = \lim_{n\to\infty} \min_{|w|=n} \|M_w\|^{1/n}.$$

For a word $w = w_1 \cdots w_k$ we write $M_w = M_{w_k} \cdots M_{w_1}$, so that the first letter acts first. The set of products of a given length, and hence $\check{\rho}$, does not depend on this convention; for instance, in [9] we multiply in the other order.

For the Metzler family $\mathcal{A}$ of Section 2, set $\mathcal{B}_h = \{I + hA_i\}$ and $\mathcal{R}_h = \{(I - hA_i)^{-1}\}$ on the admissible step ranges specified below. Their associated rates are

$$g(h) = (\check{\rho}(\mathcal{B}_h) - 1)/h, \qquad G(h) = (1 - \check{\rho}(\mathcal{R}_h)^{-1})/h.$$

Under the hypotheses of Corollary 4.1, the fixed-antinorm Gripenberg algorithm [9] approximates these radii to prescribed accuracy.

The explicit bound follows by expressing each flow as a nonnegative sum of Euler iterates and transferring their antinorm growth inequality to the flow.

**Theorem 3.1** (Explicit Euler lower bound)**.** *Let $0 < h < h_0 := 1/c_A$, with $h_0 = +\infty$ when $c_A = 0$, and define $g(h) := \big(\check{\rho}(\mathcal{B}_h) - 1\big)/h$. Then, the following properties hold:*

(i) *$g(h) \le \check{\lambda}$ and $g$ is nonincreasing in $h$ (i.e., smaller steps give better bounds);*

(ii) *$\check{\lambda}_{\min} \le \lim_{h\downarrow 0} g(h) \le \check{\lambda}$, with $\check{\lambda}_{\min}$ as in Section 2. In particular $g(h) \to \check{\lambda}$ when $\check{\lambda}_{\min} = \check{\lambda}$, for instance when every $A_i$ is irreducible (Lemma 2.4);*

(iii) *if $m = 1$, then $g(h) = \eta(A_1)$ for every such $h$.*

*Proof.* Write $B_i = I + hA_i$. These matrices are nonnegative with positive diagonal, so the lower spectral radius is positive:

$$\rho^* := \check{\rho}(\mathcal{B}_h) > 0.$$

Define

$$a(x) = \inf_w \frac{\mathbf{1}^\top B_w x}{(\rho^*)^{|w|}}, \qquad x \in K.$$

As the infimum of linear functionals, $a$ is concave, positively homogeneous, superadditive and monotone on $K$. In addition,

$$a(x) \le \mathbf{1}^\top x, \quad a(B_i x) \ge \rho^* a(x), \quad a(\mathbf{1}) \ge 1,$$

where the last property follows from $\mathbf{1}^\top B_w \mathbf{1} \ge \|B_w\|_1 \ge (\rho^*)^{|w|}$ by Fekete's lemma. By uniformization, $e^{tA_i} = e^{-t/h}\sum_k \frac{(t/h)^k}{k!} B_i^k$, a series of nonnegative matrices. Superadditivity and monotonicity therefore give

$$a(e^{tA_i}x) \ge e^{-t/h}\sum_k \frac{(t\rho^*/h)^k}{k!} a(x) = e^{t(\rho^*-1)/h} a(x).$$

Composing along any switching signal yields

$$d\|\Phi_\sigma(t)\|_1 \ge \mathbf{1}^\top \Phi_\sigma(t)\mathbf{1} \ge a(\Phi_\sigma(t)\mathbf{1}) \ge e^{tg(h)}.$$

Now let $h' = \theta h$ with $\theta \in (0,1)$. Then $I + h'A_i = (1-\theta)I + \theta B_i$, and the same $a$ satisfies $a((I+h'A_i)x) \ge (1-\theta+\theta\rho^*)a(x)$. Now use the following fact: if $a(B_i' x) \ge \gamma a(x)$ for all $i$ and $x$ and some $\gamma > 0$, then $\check{\rho}(\{B_i'\}) \ge \gamma$. Indeed, $d\|B_w'\|_1 \ge a(B_w'\mathbf{1}) \ge \gamma^{|w|}$. Hence $\check{\rho}(\mathcal{B}_{h'}) - 1 \ge \theta(\rho^* - 1)$, that is, $g(h') \ge g(h)$, proving (i).

By (i), $g(h)$ increases as $h \downarrow 0$ and is bounded by $\check{\lambda}$, so the limit exists and is at most $\check{\lambda}$. Fix $\nu < \check{\lambda}_{\min}$. Lemma 2.5(i), with $C = K$, gives a concave, positively homogeneous function $a_T$ on $\mathbb{R}^d$, Lipschitz with some constant $L$ in the $\ell^1$ norm, such that:

- $a_T \ge c_0 > 0$ on $\Delta$, and $0 \le a_T(x) \le C'\mathbf{1}^\top x$ on $K$, because its coefficient vectors are nonnegative and bounded by some $C'$;
- $a_T(e^{sA_i}y) \ge e^{\nu s} a_T(y)$ for $y \in \Delta$ and $s \in [0,1]$.

Let $h \le 1$. For $y \in \Delta$, $\|(I + hA_i)y - e^{hA_i}y\|_1 \le C_1 h^2$ with $C_1 = \frac{1}{2}K_A^2 e^{K_A}$. Hence

$$a_T(B_i y) \ge e^{\nu h} a_T(y) - LC_1 h^2 \ge \gamma\, a_T(y), \qquad \gamma = e^{\nu h} - LC_1 h^2/c_0,$$

and by homogeneity the same holds for every $y \in K$. Take $h$ sufficiently small that $\gamma > 0$, which is possible since $\gamma \to 1$ as $h \downarrow 0$. Then $C'd\,\|B_w\|_1 \ge a_T(B_w\mathbf{1}) \ge \gamma^{|w|} a_T(\mathbf{1})$ for every word, with $a_T(\mathbf{1}) \ge dc_0 > 0$. Therefore

$$\check{\rho}(\mathcal{B}_h) \ge \gamma \overset{e^x - 1 \ge x}{\Longrightarrow} g(h) \ge \nu - LC_1 h/c_0,$$

Letting $h \downarrow 0$ and then $\nu \uparrow \check{\lambda}_{\min}$ gives $\liminf g \ge \check{\lambda}_{\min}$. When every mode is irreducible, $\check{\lambda}_{\min} = \check{\lambda}$ by Lemma 2.4, concluding the proof of (ii).

Finally, the last property (iii) is an immediate consequence of the fact that the Perron root of $I + hA$ is $1 + h\eta(A)$. □

*Remark* 3.2. By Theorem 2.7, $\check{\lambda}_{\min}$ is the smallest exponent of an invariant face, so $g(h) \to \check{\lambda}$ whenever every invariant face is slow. Proposition 3.7 and Corollary 3.8 give sufficient conditions for $g(h) \to \check{\lambda}$ that also cover families with $\check{\lambda}_{\min} < \check{\lambda}$; however, in general the question is open. Refer to Section 5 for a more thorough discussion.

**Lemma 3.3.** *Let $T_0 > 0$. There is $C$, depending only on $m$, $K_A$ and $T_0$, with the following property. Let $0 < h \le 1$ with $hK_A \le \frac{1}{2}$, let $q \ge 1$ be an integer with $qh \le T_0$, and let $\sigma$ be a switching signal on $[0, qh]$. Then there is a word $w$ of length $q$ with*

$$\|R_w - \Phi_\sigma(qh)\|_1 \le Ch, \qquad \text{where } R_i = (I - hA_i)^{-1}.$$

*Proof.* Let $U_i(t)$ be the time that $\sigma$ spends in mode $i$ up to $t$ and choose $w = w_1 \cdots w_q$ so that $w_k$ is a mode maximizing

$$U_i(kh) - W_i((k-1)h), \qquad \text{where } W_i(t) := |\{s \le t \ : \ w(s) = i\}|.$$

Here $w(s) = w_k$ for $s \in [(k-1)h, kh)$ is the piecewise-constant signal of the word. At grid times, $W_i((k-1)h)$ is $h$ times the number of letters $i$ among $w_1, \dots, w_{k-1}$. For simplicity's sake, we split the proof into several small steps:

- **Discrepancy.** The deficits $U_i - W_i$ at grid times sum to 0. A deficit decreases only when its mode is chosen, and the chosen mode has deficit at least $h/m$ before the step. So every deficit at grid times lies in $(-h, (m-1)h]$. Between grid times $U_i$ and $W_i$ are nondecreasing and grow by at most $h$, so $D_i = W_i - U_i$ satisfies $|D_i(t)| \le mh$ for all $t$ (and $D_i \equiv 0$ if $m = 1$). Both $W_i$ and $U_i$ are Lipschitz, and $\mathrm{d}D_i = (\mathbf{1}[w(s) = i] - \mathbf{1}[\sigma(s) = i])\,\mathrm{d}s$.

- **Exponential words.** Let $\Phi^w$ be the transition matrix of the signal $w$. For $x(t) = \Phi_\sigma(t)x_0$, variation of constants gives

$$\Phi^w(t)x_0 - x(t) = \sum_i \int_0^t \Phi^w(t,s) A_i x(s)\,\mathrm{d}D_i(s).$$

  Integrate by parts, using $D_i(0) = 0$, $|D_i| \le mh$, $\partial_s \Phi^w(t,s) = -\Phi^w(t,s)A_{w(s)}$ and $\dot{x} = A_\sigma x$. This gives $\|\Phi^w(qh) - \Phi_\sigma(qh)\|_1 \le C_4 h$, with $C_4 := C_4(m, K_A, T_0)$.

- **Resolvent words.** For $hK_A \le 1/2$ the Neumann series gives $\|R_i\|_1 \le e^{2hK_A}$ and

$$\|R_i - e^{hA_i}\|_1 \le \sum_{k \ge 2} (hK_A)^k \le 2K_A^2 h^2.$$

  A telescoping sum over the $q \le T_0/h$ factors gives $\|R_w - \Phi^w(qh)\|_1 \le 2K_A^2 T_0 e^{2K_A T_0} h$.

So the lemma holds with $C := C_4 + 2K_A^2 T_0 e^{2K_A T_0}$. □

For implicit Euler, instead, the resolvent averages the flow over an exponential waiting time. This representation gives the upper bound. Lemma 3.3 supplies the approximation needed for convergence.

**Theorem 3.4** (Implicit Euler upper bound)**.** *Let $h > 0$ satisfy $h\,\eta(A_i) < 1$ for all $i$, and define the rate $G(h) = \big(1 - \check{\rho}(\mathcal{R}_h)^{-1}\big)/h$. Then:*

(i) *$\check{\lambda} \le G(h)$. In particular, every word $w$ of length $k \ge 1$ gives the upper bound*

$$\check{\lambda} \le \frac{1 - \rho(R_w)^{-1/k}}{h};$$

(ii) *$G$ is nondecreasing in $h$;*

(iii) *$G(h) \to \check{\lambda}$ as $h \downarrow 0$;*

(iv) *if $m = 1$, then $G(h) = \eta(A_1)$.*

*Proof.* (i) Since $h\eta(A) < 1$, the resolvent can be written in integral form as

$$(I - hA)^{-1} = \int_0^\infty e^{-s} e^{shA} \, \mathrm{d}s.$$

For independent $S_1, \dots, S_k \sim \mathrm{Exp}(1)$, it follows that $R_w = \mathbb{E}\, \Phi_\sigma(T)$, where $\sigma$ uses the letters of $w$ for the durations $hS_j$ and $T = h\sum_j S_j \sim h\,\Gamma(k,1)$. By Proposition 2.2(i), $\|\Phi_\sigma(T)\|_1 \geq e^{\check{\lambda}T}$. Moreover $h\check{\lambda} \leq h \min_i \eta(A_i) < 1$. Therefore

$$d\|R_w\|_1 \geq \mathbb{E}\, \mathbf{1}^\top \Phi_\sigma(T)\mathbf{1} \geq \mathbb{E}\, e^{\check{\lambda}T} = (1 - h\check{\lambda})^{-k}.$$

Hence $\check{\rho}(\mathcal{R}_h) \geq (1 - h\check{\lambda})^{-1}$. The word bound follows from $\check{\rho} \leq \rho(R_w)^{1/k}$.

(ii) Let $0 < h' < h$ and $\theta = h'/h \in (0,1)$. Write $R_i = (I - hA_i)^{-1}$, $R_i' = (I - h'A_i)^{-1}$ and $\gamma = \check{\rho}(\mathcal{R}_{h'})$. Then, since $I - hA_i = \theta^{-1}\big((I - h'A_i) - (1-\theta)I\big)$, we have

$$R_i = \theta R_i' \big(I - (1-\theta)R_i'\big)^{-1} = \theta \sum_{k \geq 0} (1-\theta)^k (R_i')^{k+1}.$$

The series converges. By the proof of (iv), $\rho(R_i') = (1 - h'\eta(A_i))^{-1}$, and $(1-\theta) < 1 - h'\eta(A_i)$ because $h\eta(A_i) < 1$. For the same reason $(1-\theta)\gamma < 1$, since $\gamma \leq \min_i \rho(R_i')$.

Set $a(x) := \inf_w \mathbf{1}^\top R_w' x / \gamma^{|w|}$ for $x \in K$. As in the proof of Theorem 3.1(i), $a$ is concave, positively homogeneous, superadditive and monotone on $K$. It satisfies $a(x) \leq \mathbf{1}^\top x$, $a(\mathbf{1}) \geq 1$ and $a(R_i' x) \geq \gamma a(x)$. Superadditivity extends to convergent series $\sum_k y_k$ with $y_k \in K$:

$$a\left(\sum_k y_k\right) \geq a\left(\sum_{k \leq n} y_k\right) \geq \sum_{k \leq n} a(y_k) \qquad \text{for every } n \in \mathbb{N}.$$

Hence

$$\begin{aligned} a(R_i x) &\geq \theta \sum_{k \geq 0} (1-\theta)^k a\big((R_i')^{k+1} x\big) \\ &\geq \theta \sum_{k \geq 0} (1-\theta)^k \gamma^{k+1} a(x) = \beta\, a(x), \qquad \beta := \frac{\theta\gamma}{1 - (1-\theta)\gamma}. \end{aligned}$$

By the fact used in the proof of Theorem 3.1(i), $\check{\rho}(\mathcal{R}_h) \geq \beta$. Then

$$\check{\rho}(\mathcal{R}_h)^{-1} \leq \beta^{-1} = (\theta\gamma)^{-1} - (1-\theta)/\theta \implies 1 - \check{\rho}(\mathcal{R}_h)^{-1} \geq \theta^{-1}(1 - \gamma^{-1}).$$

Dividing by $h$ gives $G(h) \geq (1 - \gamma^{-1})/h' = G(h')$.

(iii) By (i) and (ii), $\lim_{h \downarrow 0} G(h)$ exists and is at least $\check{\lambda}$. Fix $\varepsilon > 0$. By Proposition 2.2(i) there are $T > 0$ and a switching signal $\sigma$ with $\|\Phi_\sigma(T)\|_1 \leq e^{(\check{\lambda}+\varepsilon)T}$. Take $h = T/N$ with $N$ large. Once $h \leq 1$ and $hK_A \leq 1/2$, Lemma 3.3 with $T_0 = T$ and $q = N$ gives a word $w$ of length $N$ with $\|R_w - \Phi_\sigma(T)\|_1 \leq C_T h$, where $C_T$ depends only on $m$, $K_A$ and $T$. By the word bound in (i) and the inequality $1 - e^{-y} \leq y$,

$$G(T/N) \leq \frac{1 - \rho(R_w)^{-1/N}}{h} \leq \frac{\log \rho(R_w)}{Nh} \leq \frac{1}{T} \log\big(\|\Phi_\sigma(T)\|_1 + C_T h\big).$$

Letting $N \to \infty$ gives $\lim_{h \downarrow 0} G(h) \leq \check{\lambda} + \varepsilon$. Since $\varepsilon$ is arbitrary, the limit is $\check{\lambda}$.

(iv) The eigenvalues of $(I - hA)^{-1}$ are $(1 - h\mu)^{-1}$ for $\mu$ in the spectrum of $A$. Write $A = N - c_A I$ with $N \geq 0$. Then $|\mu + c_A| \leq \rho(N) = \eta(A) + c_A$, so

$$|1 - h\mu| \geq 1 + hc_A - h|\mu + c_A| \geq 1 - h\eta(A) > 0,$$

with equality at $\mu = \eta(A)$. Hence $\rho\big((I - hA)^{-1}\big) = (1 - h\eta(A))^{-1}$. □

### 3.2 First-order convergence

For irreducible modes, local $O(h^2)$ errors measured by a Lipschitz antinorm give the explicit estimate. Uniform positivity converts the resolvent approximation on fixed time blocks into multiplicative bounds, yielding the implicit estimate.

**Theorem 3.5.** *Suppose every $A_i$ is irreducible. There are $h_1 > 0$ and $C > 0$ such that, for $0 < h \le h_1$,*

$$\check{\lambda} - Ch \le g(h) \le \check{\lambda} \le G(h) \le \check{\lambda} + Ch.$$

*Proof.* Fix $\tau > 0$. We take $h_1$ so small that every $h \in (0, h_1]$ satisfies:

- $h < h_0$ and $h\,\eta(A_i) < 1$ for all $i$, so that $g(h)$ and $G(h)$ are defined;
- $h \le \min(1, \tau)$; $hK_A \le \frac{1}{2}$; $2C_1h^2 < \rho'$;
- $C_2h^2 < e^{\check{\lambda}h}$, with $C_1$, $C_2$ and $\rho'$ as below.

The two inner inequalities are Theorem 3.1(i) and Theorem 3.4(i). Now, let $a$ be the function provided by Lemma 2.5(ii). We recall that it satisfies:

- $a(e^{sA_i}x) \ge e^{\check{\lambda}s}a(x)$ for all $x \in K$ and $s \ge 0$;
- $c_a\mathbf{1}^\top x \le a(x) \le C_a\mathbf{1}^\top x$ on $K$, for some constants $0 < c_a \le C_a$;
- $a$ is Lipschitz, with some constant $L$, on an $\ell^1$-neighbourhood $N'$ of $\Delta$ of radius $\rho' > 0$.

Let $y \in \Delta$, $w = e^{hA_i}y$, $z = (I + hA_i)y$ and $s = \mathbf{1}^\top w \in [e^{-K_Ah}, e^{K_Ah}]$. Then, for $h \le 1$,

$$\|z - w\|_1 \le C_1h^2, \qquad C_1 := \tfrac{1}{2}K_A^2e^{K_A}.$$

So $\|z/s - w/s\|_1 \le 2C_1h^2 < \rho'$, and $z/s \in N'$. By homogeneity, $|a(z) - a(w)| = s\,|a(z/s) - a(w/s)| \le LC_1h^2$. Hence

$$a\big((I + hA_i)y\big) \ge e^{\check{\lambda}h}a(y) - LC_1h^2 \ge \big(e^{\check{\lambda}h} - C_2h^2\big)a(y), \qquad C_2 := LC_1/c_a,$$

and by homogeneity the same holds for every $y \in K$. Set $\gamma = e^{\check{\lambda}h} - C_2h^2 > 0$. As in the proof of Theorem 3.1(i), $C_a d\,\|B_w\|_1 \ge a(B_w\mathbf{1}) \ge \gamma^{|w|}a(\mathbf{1})$ for every word, so

$$\check{\rho}(\mathcal{B}_h) \ge \gamma \implies g(h) \ge \frac{e^{\check{\lambda}h} - 1}{h} - C_2h \ge \check{\lambda} - C_2h.$$

By Lemma 2.4 and $\Phi_\sigma(t, \tau) \ge e^{-c_A(t-\tau)}I$, there is $\delta_* > 0$ with $\Phi_\sigma(t) \ge \delta_*\mathbf{1}\mathbf{1}^\top$ for every switching signal and every $t \in [\tau, 2\tau]$. Let $q = \lceil \tau/h \rceil$, so that $qh \in [\tau, 2\tau)$. By Lemma 3.3 with $T_0 = 2\tau$, every switching signal $\sigma$ on $[0, qh]$ has a word $w$ of length $q$ with $\|R_w - \Phi_\sigma(qh)\|_1 \le C'h$. So every entry of $R_w - \Phi_\sigma(qh)$ is at most $C'h$, and since $\Phi_\sigma(qh) \ge \delta_*\mathbf{1}\mathbf{1}^\top$, we get $0 \le R_w \le (1 + C_3h)\,\Phi_\sigma(qh)$ entrywise, with $C_3 := C'/\delta_*$.

For $N \ge 1$ and $T = Nqh$, choose $\sigma$ with $\|\Phi_\sigma(T)\|_1 \le 2\bar{\mu}(T)$ and apply the block construction to each block. Products of nonnegative matrices preserve entrywise inequalities, so the resulting word $w$ of length $Nq$ satisfies $R_w \le (1 + C_3h)^N\Phi_\sigma(T)$. By Fekete's lemma $\check{\rho}(\mathcal{R}_h) \le \|R_w\|_1^{1/(Nq)}$, and $Nq = T/h$ with $T \ge N\tau$, so

$$\log\check{\rho}(\mathcal{R}_h) \le \frac{h}{T}\Big(N\log(1 + C_3h) + \log\big(2\bar{\mu}(T)\big)\Big) \le \frac{C_3}{\tau}h^2 + \frac{h}{T}\log\big(2\bar{\mu}(T)\big).$$

Letting $N \to \infty$ gives $\log\check{\rho}(\mathcal{R}_h) \le h(\check{\lambda} + C_3h/\tau)$. Since $1 - e^{-y} \le y$, we finally conclude that

$$G(h) \le \check{\lambda} + (C_3/\tau)h.$$

□

*Remark* 3.6. The order is **sharp**. In Example 3.10, both errors equal, including for irreducible modes,

$$(1-\sqrt{1-h^2})/h = h/2 + O(h^3),$$

and under Theorem 3.12 the explicit error, when chattering is optimal, is at least

$$Qh/2 - O(h^2).$$

The constant $C_2 = LC_1/c_a$, with $c_a = (d\kappa_0)^{-1}$, $L = 2\kappa_0(C_0'+1)$ and $\kappa_0 = M/\delta$ from Lemma 2.5(ii), can exceed the observed widths $0.02h$–$1.5h$ (Table 2); Theorem 3.12 refines it. Without irreducibility, implicit convergence remains general, without a rate. Norms absorb the additive errors in its proof, whereas the lower argument needs a flow-invariant function bounded above and below on $\Delta$, capped at $\check{\lambda}_{\min}$ (Lemma 2.3(ii)).

### 3.3 Invariant cones and reducible families

Without irreducibility, the general estimate only places the explicit limit between $\check{\lambda}_{\min}$ and $\check{\lambda}$. The next criteria establish convergence when the relevant invariant faces are slow, using cones that the Euler matrices preserve.

**Proposition 3.7** (Invariant cones)**.** *Let $C \subseteq K$ be a closed convex cone, $C \neq \{0\}$, and assume that $(I + h_C A_i)C \subseteq C$ for all $i$ and some $h_C \in (0, h_0)$. Then $C$ is invariant under every flow $e^{tA_i}$, so $\check{\lambda}_C$ is defined (Section 2). In addition:*

(i) $\lim_{h\downarrow 0} g(h) \geq \check{\lambda}_C$.

(ii) *If $C \setminus \{0\} \subset \operatorname{int} K$, then $\check{\lambda}_C \geq \check{\lambda}$, so $g(h) \to \check{\lambda}$.*

(iii) *Suppose that, for some $x_0 > 0$, the vectors $B_w x_0/\mathbf{1}^\top B_w x_0$, with $w$ ranging over all words of $\mathcal{B}_{h_C}$, stay in a compact subset of $\operatorname{int}\Delta$. Then the closed convex cone they generate satisfies the hypotheses of* (ii).

(iv) *Replacing every $A_i$ by $A_i^\top$ changes neither $g$ nor $\check{\lambda}$, so* (i)–(iii) *may be applied to the transposed family.*

*A cone as in* (ii) *exists only if every $A_i$ has a positive eigenvector. Analogously, for the transposed family, only if every $A_i$ has a positive left eigenvector.*

*Proof.* For $h = \theta h_C$ with $\theta \in (0,1]$, $I + hA_i = (1-\theta)I + \theta(I + h_C A_i)$ maps $C$ into $C$. Since $e^{tA_i} = \lim_n (I + (t/n)A_i)^n$ and $C$ is closed, every $e^{tA_i}$ maps $C$ into $C$, and hence so does every $\Phi_\sigma(t)$ (Section 2). We now divide the proof into the four statements:

(i) Repeat the proof of Theorem 3.1(ii) with $\Delta$ replaced by $C \cap \Delta$. Lemma 2.5(i) gives, for every $\nu < \check{\lambda}_C$, a function $a_T$ with the properties listed there. For sufficiently small $0 < h \leq h_C$ we have $\gamma > 0$. Since $B_i y \in C$, the comparison with $e^{hA_i}y$ gives $a_T(B_i y) \geq \gamma a_T(y)$ on $C$; iteration from any $x_0 \in C \setminus \{0\}$ gives $\check{\rho}(\mathcal{B}_h) \geq \gamma$.

(ii) By compactness, $\varepsilon := \min\{x_j \,:\, x \in C \cap \Delta,\ j \leq d\}$ is strictly positive. For $x \in C \cap \Delta$, $\mathbf{1}^\top \Phi_\sigma(t)x \geq \varepsilon\, \mathbf{1}^\top \Phi_\sigma(t)\mathbf{1} \geq \varepsilon\, \bar{\mu}(t) \geq \varepsilon e^{\check{\lambda}t}$ by Proposition 2.2(i). So $\check{\lambda}_C \geq \check{\lambda}$, and (i) together with Theorem 3.1(i) gives $g(h) \to \check{\lambda}$.

(iii) Each $B_i$ maps the ray of each of these vectors onto the ray of another one, so it maps the closed convex cone they generate into itself. If the vectors lie in a compact set $X_0 \subset \operatorname{int}\Delta$, the base of that cone, the closed convex hull of the vectors, lies in $\operatorname{conv} X_0$, which is compact and contained in $\operatorname{int}\Delta$.

(iv) If $\tilde{\sigma}(s) = \sigma(t-s)$, the transition matrix of the transposed family is $\Phi_{\tilde{\sigma}}(t)^\top$, and products of the matrices $I + hA_i^\top$ are transposes of products of the $B_i$ in reverse order. Since $\|M^\top\|_1 \leq d\,\|M\|_1$, the exponential rates are unchanged.

The fixed-point argument in the proof of Theorem 2.7 gives a rest point of the projective flow of each $A_i$ in $C \cap \Delta$, that is,

$$A_i v = \mu v \quad \text{with } v \in C \setminus \{0\}.$$

Under (ii), $v > 0$. If $u \geq 0$, $u \neq 0$, is a left eigenvector of $A_i$ for $\eta(A_i)$, then $\eta(A_i)u^\top v = u^\top A_i v = \mu u^\top v$ with $u^\top v > 0$, so $\mu = \eta(A_i)$. □

**Corollary 3.8.** *Let $C$ be as in Proposition 3.7. Then*

$$\liminf_{h \downarrow 0} g(h) \;\geq\; \min\{\check{\lambda}_J \;:\; J \in \Sigma(C),\ S(J) \neq \emptyset\}.$$

*In particular, if every face $F_J$ with $J \in \Sigma(C)$ is slow, then $g(h) \to \check{\lambda}$.*

*Proof.* Combine Proposition 3.7(i) and Theorem 2.7. If every face in the (nonempty) minimum is slow, the minimum is at least $\check{\lambda}$, so $\check{\lambda} \geq \liminf g(h) \geq \check{\lambda}$ by Theorem 3.1(i). □

Proposition 3.7(ii) corresponds to $\Sigma(C) = \{\{1, \dots, d\}\}$; the corollary also permits contact with slow boundary faces. For $C = K$, it recovers Theorem 3.1(ii) through Theorem 2.7, giving convergence when all invariant faces are slow.

An Euler-invariant cone containing $x > 0$ contains the limits of $B_i^n x/(\mathbf{1}^\top B_i^n x)$, including the dominant eigenvector when unique. If that vector lies on a fast face, the criterion cannot apply and convergence remains undecided. A cone inside $\partial K$ lies in a common invariant face and helps only when that face has exponent $\check{\lambda}$.

**Computation 3.9.** For cases 77, 148, 677, 1195 and 1440 (Appendix A), only the last has two irreducible modes. In both orientations, every proper invariant face has $S(J) = \{i\}$. Its exponent $\eta(A_i[J, J])$ is bounded below exactly by Collatz–Wielandt with a nonnegative eigenvector. We prove slowness by comparison with a certified upper bound for $c = \min_\alpha \eta(S_\alpha) \geq \check{\lambda}$, using Proposition 2.2(iii) at rational $\alpha$ near the minimizer.

- *All invariant faces slow.* In case 1195 as given, the only invariant proper face is $F_{\{2,3\}}$, invariant under $A_1$, with exponent at least $-0.289318$, while $c \leq -0.532919$.

  In case 677 as given it is $F_{\{2\}}$, invariant under $A_2$, with exponent $-0.237097$, while $c \leq -0.394492$.

  In case 77 after transposition it is $F_{\{2\}}$, invariant under $A_2^\top$, with exponent 0.024892, while $c \leq -0.148186$. Case 1440 has none. By Theorem 2.7, $\check{\lambda}_{\min} = \check{\lambda}$ for these families, and Theorem 3.1(ii) gives $g(h) \to \check{\lambda}$.

  The orientation matters, although $g$ and $\check{\lambda}$ do not depend on it (Proposition 3.7(iv)): case 77 as given and cases 677 and 1195 transposed each have an invariant face with exponent below $c$, which this comparison cannot show to be slow. For cases 77 and 677 in these orientations, the invariant polygons of Proposition 3.7(ii) (with $h_C = 0.2$, of 21 and 53 vertices, coordinates at least 0.0796 and 0.1137) give the same conclusion.

- In case 148 both orientations have a fast face (Example 2.8), so we use Corollary 3.8 for the transposed family, with $h_C = 1/5 < h_0$ (Figure 1(b)). Its other invariant face, $F_{\{1,3\}}$ under $A_2^\top$, has exponent at least 0.147542, while $c \leq -1.236979$, so it is slow. The cone

  $$C = \{x \geq 0 \;:\; (1-\beta)(x_1 + x_2) \geq \beta x_3\}, \qquad \beta = 1/20,$$

  has extreme rays $e_1$, $e_2$, $(\beta, 0, 1-\beta)$ and $(0, \beta, 1-\beta)$, and cuts off only $e_3$. The supports of its points are all nonempty subsets of $\{1, 2, 3\}$ except $\{3\}$, and the only proper invariant face among them is the slow face $F_{\{1,3\}}$. Both Euler matrices map the four extreme rays into $C$. This is checked in exact rational arithmetic, with binary64 data, and replayed by an independent method (barycentric coordinates in a fan triangulation). By Corollary 3.8, $g(h) \to \check{\lambda}$. So $g(h) \to \check{\lambda}$ for all five test pairs.

- *Case 148 without a cone.* In the given orientation $\check{\lambda}_{\min} = (A_2)_{22}$ (Example 2.8). The polytopes of Remark 4.5, which have no vertex on the axis of $e_2$, certify $\lim_{h\downarrow 0} g(h) \geq -1.247834 > \check{\lambda}_{\min}$ directly.

### 3.4 Sharpness and refined coefficients

We first give an exact example showing that the linear rate is sharp, then derive one-sided coefficients for two-mode families.

**Example 3.10.** Let $\varepsilon \geq 0$ and $A_{1,2} = \left(\begin{smallmatrix} \mp 1 & \varepsilon \\ \varepsilon & \pm 1 \end{smallmatrix}\right)$. For $\varepsilon = 0$ the modes are diagonal; for $\varepsilon > 0$ they are irreducible. For $x > 0$, $a(x) = \sqrt{x_1 x_2}$ satisfies, along both modes,

$$\frac{\mathrm{d}}{\mathrm{d}t} \log a(x(t)) = \frac{\varepsilon}{2}\Big(\frac{x_2}{x_1} + \frac{x_1}{x_2}\Big) \geq \varepsilon,$$

so $\check{\lambda} \geq \varepsilon$. Since $\eta(S_{1/2}) = \varepsilon$, Proposition 2.2(iii) gives $\check{\lambda} = \varepsilon$. Since $\eta(A_i) = \sqrt{1+\varepsilon^2}$, let $0 < h < (1+\varepsilon^2)^{-1/2}$, so that $h < h_0 = 1$ and $h\,\eta(A_i) < 1$, and put $r = \sqrt{1-h^2} + \varepsilon h$.

- *Explicit family.* Let $a(x) = \sqrt{x_1 x_2}$ and $t = x_2/x_1$. By AM–GM,

$$\frac{a(B_1 x)^2}{a(x)^2} = \frac{(1-h)h\varepsilon}{t} + 1 - h^2 + h^2\varepsilon^2 + h\varepsilon(1+h)\,t \;\geq\; r^2,$$

  and the same holds for $B_2$. Since $a(x) \leq \frac{1}{2}\mathbf{1}^\top x$ and $a(\mathbf{1}) = 1$, the fact used in the proof of Theorem 3.1(i) gives $\check{\rho}(\mathcal{B}_h) \geq r$. Conversely, $\check{\rho}(\mathcal{B}_h) \leq \rho(B_1 B_2)^{1/2}$ and

$$\rho(B_1 B_2) = 1 - h^2(1-\varepsilon^2) + 2h\varepsilon\sqrt{1-h^2} = r^2 \implies \check{\rho}(\mathcal{B}_h) = r.$$

- *Implicit family.* For a $2\times 2$ matrix $A$ with $\operatorname{tr} A = 0$, $\operatorname{adj}(I - hA) = I + hA$. So $(I - hA_i)^{-1} = B_i/D$ with $D = \det(I - hA_i) = 1 - h^2(1+\varepsilon^2) = (\sqrt{1-h^2} - \varepsilon h)\,r$, the same for both modes. Hence $\check{\rho}(\mathcal{R}_h) = r/D$.

Therefore

$$g(h) = \check{\lambda} - \frac{1-\sqrt{1-h^2}}{h}, \quad G(h) = \check{\lambda} + \frac{1-\sqrt{1-h^2}}{h}, \quad \frac{1-\sqrt{1-h^2}}{h} = \frac{h}{2} + O(h^3).$$

For every $\varepsilon \geq 0$, and in particular for irreducible modes, both one-sided errors are of exact order $h$, and the midpoint $(g+G)/2$ equals $\check{\lambda}$.

*Remark* 3.11. Let $m = 2$, and suppose $\eta = \eta(S_\alpha)$ is a simple eigenvalue with left and right eigenvectors $u, v$. Writing $F = A_1 - A_2$, one has $[A_1, A_2] = [F, S_\alpha]$ and hence

$$u^\top [A_1, A_2] v = \eta\, u^\top F v - \eta\, u^\top F v = 0.$$

At $\alpha = 1/2$, under the same simplicity assumption, Baker–Campbell–Hausdorff gives

$$\log \rho(e^{hA_2} e^{hA_1})/(2h) = \eta(S_{1/2}) + O(h^2).$$

The closest statement is [3, Prop. 10]: periodic perturbations of the best constant control change the Floquet eigenvalue only at second order. For general noncommuting matrices, splitting formulas with positive durations have order at most two [20, 21]. Note that special families, such as commuting ones, can be exact.

We next derive explicit one-sided first-order bounds for the Euler errors at a two-mode chattering optimum.

**Theorem 3.12.** *Let $m = 2$, $F = A_1 - A_2$ and $S_a = aA_1 + (1-a)A_2$ for $a \in [0,1]$. Suppose that $a^* \in (0,1)$ minimizes $\eta(S_a)$ over $[0,1]$ and that $S = S_{a^*}$ is irreducible. Let $c = \eta(S)$, $u^\top S = cu^\top$, $Sv = cv$, $u^\top v = 1$, and*

$$Q = a^*(1-a^*)\,u^\top F^2 v = u^\top\big(a^*A_1^2 + (1-a^*)A_2^2\big)v - c^2.$$

*There are $h_2 > 0$ and $C > 0$ such that, for $0 < h \le h_2$,*

$$g(h) \le c - \tfrac{Q}{2}h + Ch^2, \qquad G(h) \le c + \tfrac{Q}{2}h + Ch^2.$$

*If moreover chattering is optimal, $\check\lambda = c$, as for every planar pair [1, Thm. 1], then*

$$\check\lambda - g(h) \ge \tfrac{Q}{2}h - Ch^2, \qquad 0 \le G(h) - \check\lambda \le \tfrac{Q}{2}h + Ch^2,$$

*and $Q \ge 0$. For $d = 2$, $Q = \frac{1}{2}a^*(1-a^*)(c-\lambda_2)\,\eta''(a^*)$, where $\lambda_2$ is the other eigenvalue of $S$ and $\eta(a) = \eta(S_a)$. Hence, $Q > 0$ when the minimum is nondegenerate.*

*Proof.* Every $S_a$, $0 < a < 1$, has the off-diagonal pattern of $S$, so $\eta(a)$ is a simple eigenvalue of $S_a$, analytic in $a$, with Perron vectors $u_a, v_a$ ($u_a^\top v_a = 1$) and $\eta'(a^*) = u^\top F v = 0$. Set

$$\kappa(a) = u_a^\top(aA_1^2 + (1-a)A_2^2)v_a, \qquad Q(a) = \kappa(a) - \eta(a)^2,$$

so that $Q = Q(a^*)$. For $1 \le p < q$ let $w(p,q) = w_1 \cdots w_q$ be such that

$$w_j = \begin{cases} 1 & \text{if } \lfloor jp/q \rfloor > \lfloor (j-1)p/q \rfloor, \\ 2 & \text{otherwise.} \end{cases} \tag{3}$$

Fix a compact subinterval $\mathcal{I} \subset (0,1)$ with $a^* \in \operatorname{int}\mathcal{I}$. Lemma D.1 in Appendix D shows that, uniformly for $b \in \mathcal{I}$ and $T = qh \le T_1$, with $T_1$ fixed,

$$\frac{\log\rho(B_w)}{T} = \eta(b) - \frac{h}{2}\kappa(b) + O(h^2), \qquad \frac{\log\rho(R_w)}{T} = \eta(b) + \frac{h}{2}\kappa(b) + O(h^2).$$

Take $q = \lfloor T_1/h \rfloor$ and $p$ nearest to $qa^*$. Reduce $h_2$ so that $h \le T_1/2$, $1 \le p < q$ and $b \in \mathcal{I}$; then $|b - a^*| \le h/T_1$. With $\zeta = \log\rho(B_w)/T$ we have $\check\rho(\mathcal{B}_h) \le \rho(B_w)^{1/q} = e^{h\zeta}$, so

$$g(h) \le \frac{e^{h\zeta} - 1}{h} \le \zeta + \frac{h}{2}\zeta^2 + O(h^2) = \eta(b) - \frac{h}{2}Q(b) + O(h^2).$$

Since $\eta'(a^*) = 0$, $\eta(b) = c + O(h^2)$, and $Q(b) = Q + O(h)$. The implicit side is the same. Indeed, by applying Theorem 3.4(i) we get

$$G(h) \le (1 - e^{-h\zeta'})/h \le \zeta' - \frac{h}{2}\zeta'^2 + O(h^2), \qquad \zeta' = \log\rho(R_w)/T.$$

When $\check\lambda = c$, the second pair of inequalities in the statement follows from these bounds and $g(h) \le \check\lambda \le G(h)$. Dividing the upper bound for $G(h) - \check\lambda$ by $h$ and letting $h \downarrow 0$ gives $Q \ge 0$. For $d = 2$, write $I = vu^\top + v_2u_2^\top$ with the eigenvectors of $\lambda_2$. Here

$$u^\top F v = 0, \qquad u^\top F^2 v = (u^\top F v_2)(u_2^\top F v),$$

and second-order perturbation gives $\eta''(a^*) = 2(u^\top F v_2)(u_2^\top F v)/(c - \lambda_2)$. □

When $\check\lambda = c$ and $Q > 0$, the explicit error is at least $Qh/2 - O(h^2)$; for irreducible modes, Theorem 3.5 makes it of order $h$ (as in Example 3.10, with $Q = 1$). The implicit conclusion is only $\limsup_{h\downarrow 0}(G(h) - \check\lambda)/h \le Q/2$; matching lower estimates remain open (Section 5 and Appendix C).

**Numerical observation 3.13.** We test Example 3.10 with $\varepsilon = 0.2$ and the asymmetric planar pair given by

$$A_1 = \left(\begin{smallmatrix} -2 & 0.3 \\ 0.5 & 1 \end{smallmatrix}\right), \qquad A_2 = \left(\begin{smallmatrix} 0.5 & 0.4 \\ 0.1 & -1.5 \end{smallmatrix}\right).$$

For the latter, [1, Thm. 1] gives $\check{\lambda} = \min_\alpha \eta(S_\alpha) \approx -0.1949763083$. In Table 1, the finite-word gap satisfies $G_{\mathrm{up}} - g_{\mathrm{est}} \approx hQ$, with

$$Q = u^\top \left( \sum_i \alpha_i^* (A_i - S_{\alpha^*})^2 \right) v$$

as in Theorem 3.12 (predicted slope 1.478, observed 1.470). The computed midpoint $(g_{\mathrm{est}} + G_{\mathrm{up}})/2$ suggests second-order cancellation, while the exact midpoint $(g+G)/2$ equals $\check{\lambda}$ for the symmetric pair. These finite-search data do not establish matching coefficients for $g, G$. Exponential grid schedules converge quadratically in the symmetric case; for the asymmetric pair, the word-length cap limits approximation of $\alpha^* \approx 0.3905$.

Table 1: Euler word estimates and exponential schedules, words of length at most 16. The implicit-word value $G_{\mathrm{up}}$ and the schedule rate bound $\check{\lambda}$ from above up to rounding; $g_{\mathrm{est}}$ estimates $g(h)$. For the symmetric pair, the values match Example 3.10.

| | symmetric ($\check{\lambda} = 0.2$) | | | asymmetric ($\check{\lambda} \approx -0.194976$) | | |
|---|---|---|---|---|---|---|
| $h$ | $g_{\mathrm{est}}$ | $G_{\mathrm{up}}$ | exp. schedule | $g_{\mathrm{est}}$ | $G_{\mathrm{up}}$ | exp. schedule |
| 0.2 | 0.098979 | 0.301021 | 0.201335 | −0.358479 | −0.056019 | −0.188068 |
| 0.1 | 0.149874 | 0.250126 | 0.200333 | −0.271832 | −0.123611 | −0.192983 |
| 0.05 | 0.174984 | 0.225016 | 0.200083 | −0.232157 | −0.158550 | −0.194214 |
| 0.025 | 0.187498 | 0.212502 | 0.200021 | −0.213186 | −0.176426 | −0.194521 |
| 0.0125 | 0.193750 | 0.206250 | 0.200005 | −0.203857 | −0.185482 | −0.194598 |

*Remark* 3.14. Let $\psi(\omega) = \operatorname{Re} u^\top F(c + \imath\omega - S)^{-1} F v$ for $\omega > 0$. Then (Appendix D)

$$Q = \frac{2a^*(1-a^*)}{\pi} \int_0^\infty \psi(\omega)\, \mathrm{d}\omega.$$

To derive the associated Floquet formula, let $f$ be real, bounded, measurable and $P$-periodic with zero mean, where $P = 2\pi/\omega_0$ with $\omega_0 > 0$, and set

$$\hat{f}_k = P^{-1} \int_0^P f(t) e^{-\imath k \omega_0 t}\, \mathrm{d}t.$$

Write $\langle \cdot \rangle$ for the period average. For fixed $f, P$ and small $\varepsilon$, the system $\dot{x} = (S + \varepsilon f(t) F)x$ has a Floquet solution $x(t) = e^{\lambda_\varepsilon t} p_\varepsilon(t)$, with $p_\varepsilon$ periodic and $\langle u^\top p_\varepsilon \rangle = 1$. The monodromy is analytic in $\varepsilon$ and its Perron eigenvalue at $\varepsilon = 0$ is simple, so write

$$p_\varepsilon = v + \varepsilon z + \varepsilon^2 r + o(\varepsilon^2), \qquad \lambda_\varepsilon = c + \varepsilon\lambda_1 + \varepsilon^2 \lambda_2 + o(\varepsilon^2).$$

Substitution into $\dot{p}_\varepsilon = (S + \varepsilon f F - \lambda_\varepsilon I) p_\varepsilon$ and averaging against $u^\top$ give $\lambda_1 = \langle f \rangle u^\top F v = 0$ and

$$\dot{z} = (S - cI) z + f F v, \qquad \lambda_2 = \langle f\, u^\top F z \rangle.$$

The normalization and zero mean give $\hat{z}_0 = 0$; for $k \neq 0$, $\hat{z}_k = \hat{f}_k (c + \imath k\omega_0 - S)^{-1} F v$. These inverses exist because all other eigenvalues of the irreducible Metzler matrix $S$ have real part below $c$. Parseval's identity and conjugate pairing therefore yield

$$\lambda_\varepsilon - c = 2\varepsilon^2 \sum_{k \geq 1} |\hat{f}_k|^2 \psi(k\omega_0) + o(\varepsilon^2).$$

The series is absolutely convergent. Indeed, $f \in L^2(0,P)$ and the resolvents are bounded over the nonzero integer frequencies. Thus $Q$ is a frequency average of this quadratic coefficient. If chattering is optimal, periodic convexified controls cannot improve on $c$ (by approximation with switching). Taking $f(t) = \cos(\omega_0 t)$, with $a^* + \varepsilon f(t) \in [0,1]$, gives $\psi(\omega_0) \geq 0$ for every $\omega_0 > 0$, hence $Q \geq 0$.

# 4 Rigorous brackets

In this section, we discuss lower bounds obtained from polytopes and upper bounds obtained from implicit words, mixtures or schedules. We distinguish algorithmic guarantees, exact verification and numerical performance.

## 4.1 Algorithmic guarantees

The Euler bounds turn discrete lower-spectral-radius algorithms into continuous-time brackets. For individually irreducible modes, the fixed-antinorm Gripenberg algorithm of [9] yields a convergent scheme.

**Corollary 4.1** (Continuous-time Gripenberg bounds)**.** *Suppose every $A_i$ is irreducible, and let $0 < h < h_0$ with $h\,\eta(A_i) < 1$ for all $i$.*

*(i) The families $\mathcal{B}_h$ and $\mathcal{R}_h$ are primitive. In other words, every product of $d-1$ matrices of $\mathcal{B}_h$, and every matrix of $\mathcal{R}_h$, is positive.*

*(ii) Run the fixed-antinorm Gripenberg algorithm [9, Sec. 3.1], with a fixed positive antinorm and accuracy $\delta > 0$, on $\mathcal{B}_h$ and on $\mathcal{R}_h$. Let $t_k$ be the lower bound of the first run and $s_k$ the upper bound of the second, after $k$ levels. Then, for every $k$,*

$$\frac{t_k - 1}{h} \;\leq\; \check{\lambda} \;\leq\; \frac{1 - s_k^{-1}}{h}.$$

*Write $W_k(h,\delta)$ for the width of this bracket. If $h \leq h_1$, with $h_1$ and $C$ as in Theorem 3.5, then, for each fixed $h$ and $\delta > 0$,*

$$\limsup_{k\to\infty} W_k(h,\delta) \leq 2Ch + 5\delta h^{-1}.$$

*For $\delta = \delta(h) = o(h)$, choose $k(h)$ so that $W_{k(h)}(h,\delta(h)) \leq 2Ch + 5\delta(h)/h + h$. The brackets then converge to $\check{\lambda}$ as $h \downarrow 0$; with $\delta = h^2$ their width is at most $(2C+6)h$.*

*Proof.* For $h < h_0$, the matrix $I + hA_i$ is nonnegative, has positive diagonal and has the irreducible pattern of $A_i$. The argument in the proof of Lemma 2.4 shows that every product of $d-1$ such matrices is positive. For the resolvents, write $I - hA_i = (1+hc_A)I - hN_i$ with $N_i = A_i + c_A I \geq 0$ irreducible. Since $\rho(hN_i) = h(\eta(A_i) + c_A) < 1 + hc_A$, we have

$$(I - hA_i)^{-1} = (1+hc_A)^{-1} \sum_{k\geq 0} \big(hN_i/(1+hc_A)\big)^k,$$

which is positive and proves (i). For (ii), Guglielmi and Maiale [9, Secs. 2.3 and 3.1] state that primitive families satisfy the hypotheses of their convergence theorem; for completeness we verify the cone condition. Every product $P$ is primitive, hence asymptotically rank-one, and its secondary spectral subspace is the kernel of its positive left Perron vector $u_P$. For a product of length at least $d-1$, write $P = P'M$ with $M$ one of the finitely many positive products of length $d-1$. Then $u_P^\top$ is proportional to $(u_P^\top P')M$, so the normalized vectors $u_P$ have entries bounded below uniformly. Hence the closure of the union of the secondary subspaces meets $K$

only at 0, which is the cone condition. So the main theorem applies to both families. The first run gives $t_k \le \check{\rho}(\mathcal{B}_h) \le s'_k$ and the second $t'_k \le \check{\rho}(\mathcal{R}_h) \le s_k$, with $\limsup_{k\to\infty}(s'_k - t_k) \le \delta$ and $\limsup_{k\to\infty}(s_k - t'_k) \le \delta$, with bounds held constant once the prescribed accuracy is attained. The bracket follows from Theorem 3.1(i), Theorem 3.4(i), and the monotonicity of $x \mapsto (x-1)/h$ and $x \mapsto (1-x^{-1})/h$. For the width, we have

$$t_k \ge \check{\rho}(\mathcal{B}_h) - \delta - o(1) \implies (t_k - 1)/h \ge g(h) - \delta/h - o(1) \ge \check{\lambda} - Ch - \delta/h - o(1).$$

For the upper end, $s_k \le \check{\rho}(\mathcal{R}_h) + \delta + o(1)$, and $\check{\rho}(\mathcal{R}_h) \ge (1 - h\check{\lambda})^{-1} \ge \frac{1}{2}$ by the proof of Theorem 3.4(i), since $h|\check{\lambda}| \le hK_A \le \frac{1}{2}$ by Proposition 2.2(i) and the choice of $h_1$. Finally

$$\rho^{-1} - (\rho+\delta)^{-1} \le \frac{\delta}{\rho^2} \implies \frac{1 - s_k^{-1}}{h} \le G(h) + 4\frac{\delta}{h} + o(1) \le \check{\lambda} + Ch + 4\frac{\delta}{h} + o(1).$$

□

*Remark* 4.2. Corollary 4.1 covers a fixed antinorm. The adaptive variant [9, Sec. 3.2] enlarges the unit polytope; its spectral-radius upper bounds remain valid in exact arithmetic. Its lower-bound argument assumes that enlargement cannot decrease the operator antinorm $a(P) = \min_{v\in V} a(Pv)$. This monotonicity fails in general.

## 4.2 Construction and limitations of the bounds

The next two propositions also apply to reducible families. Their bounds are valid whenever the stated inequalities hold, independently of how the candidate vertices or words were found. We use the word *certificate* for finite data whose defining inequalities can be checked exactly to prove a bound.

**Proposition 4.3** (Polytope certificates)**.** *Let $0 < h < h_0$ and $B_i = I + hA_i$, let $V \subset K \setminus \{0\}$ be finite and nonempty, and let $t > 0$. Suppose that for every $v \in V$ and every $i$ there are weights $\lambda_u \ge 0$, $u \in V$, with $\sum_u \lambda_u = 1$ and*

$$B_i v \;\ge\; t \sum_{u\in V} \lambda_u\, u. \tag{4}$$

(i) *Then $\check{\rho}(\mathcal{B}_h) \ge t$, and hence $\check{\lambda} \ge (t-1)/h$.*

(ii) *Let $0 < h' \le h$ and $\theta = h'/h$. The same set $V$ satisfies (4) for the matrices $I + h'A_i$, with $t' = 1 - \theta + \theta t$ in place of $t$. So it yields the same bound $(t'-1)/h' = (t-1)/h$ at every smaller step.*

(iii) *Replacing every $A_i$ by $A_i^\top$ does not change $\check{\rho}(\mathcal{B}_h)$, so the same inequalities may be checked for the transposed family instead.*

*Proof.* (i) Let $\mathcal{P} = \operatorname{conv} V + K$. It is convex with $\mathcal{P} + K = \mathcal{P}$. By (4) and $B_i \ge 0$, $B_i\mathcal{P} \subseteq t\mathcal{P}$, so $B_w v \in t^{|w|}\mathcal{P}$ for every word $w$ and every $v \in V$. Since $\mathbf{1}^\top y \ge c_V := \min_{u\in V} \mathbf{1}^\top u > 0$ on $\mathcal{P}$,

$$\|B_w\|_1 \ge \frac{\mathbf{1}^\top B_w v}{\mathbf{1}^\top v} \ge \frac{c_V}{\mathbf{1}^\top v}\, t^{|w|}.$$

Hence $\check{\rho}(\mathcal{B}_h) \ge t$, and $\check{\lambda} \ge (t-1)/h$ by Theorem 3.1(i). This is the lower bound of [10, Prop. 6] (on antinorms see also [11]), or the argument of Theorem 3.1(i), for the antinorm with unit ball $\mathcal{P}$. We include the proof because this antinorm need not be positive on $K$; it vanishes on the faces of $K$ that $\mathcal{P}$ does not meet, as in case 148 (Remark 4.5).

(ii) We have $0 < h' < h_0$ and

$$(I + h'A_i)v = (1-\theta)v + \theta B_i v \;\geq\; (1-\theta)v + \theta t \sum_u \lambda_u u.$$

The right-hand side is $t'$ times a convex combination of points of $V$, with weight $(1-\theta)/t'$ on $v$ and $\theta t\lambda_u/t'$ on each $u$.

(iii) Products of the matrices $I + hA_i^\top$ are transposes of products of the $B_i$ in reverse order, and $\|M^\top\|_1 \leq d\,\|M\|_1$. □

**Proposition 4.4** (Certified upper bounds)**.** *Let $h > 0$, let $w = w_1 \cdots w_k$ be a word of length $k \geq 1$, and assume that the following holds:*

(a) *for every $i$ there is $z_i > 0$ with $(I - hA_i)z_i > 0$;*

(b) *there are vectors $x_0 > 0$ and $x_1, \ldots, x_k$ with $(I - hA_{w_j})x_j \geq x_{j-1}$ for $j = 1, \ldots, k$.*

*Let $q = \max_\ell (x_k)_\ell/(x_0)_\ell$, and let $r > 0$ satisfy $r^k \geq q$. Then $h\,\eta(A_i) < 1$ for every $i$, and*

$$\check{\lambda} \leq \frac{1}{h}(1 - r^{-1}).$$

*Proof.* By (a) and the Collatz–Wielandt bound of Proposition 2.2(iii), applied to $hA_i$, we have $h\,\eta(A_i) \leq \max_\ell (hA_i z_i)_\ell/(z_i)_\ell < 1$. So, as in the proof of Theorem 3.4(i),

$$R_i = (I - hA_i)^{-1} = \int_0^\infty e^{-s} e^{shA_i}\,\mathrm{d}s \geq 0.$$

By (b), $x_j = R_{w_j}(I - hA_{w_j})x_j \geq R_{w_j} x_{j-1}$. Since every $R_i \geq 0$, induction gives $x_k \geq R_w x_0 \geq 0$. Collatz–Wielandt for the nonnegative matrix $R_w$ and $x_0 > 0$ gives $\rho(R_w) \leq \max_\ell (R_w x_0)_\ell/(x_0)_\ell \leq q \leq r^k$. Finally, Theorem 3.4(i) gives

$$\check{\lambda} \leq (1 - \rho(R_w)^{-1/k})/h \leq (1 - r^{-1})/h.$$

□

### 4.2.1 Numerical search and exact verification

Floating-point searches propose candidates, which are then checked in exact rational arithmetic and verified independently. Appendix F gives the polytope, implicit-word, mixture and schedule procedures. Its primal–dual precision concerns the chosen polytope, not $\check{\lambda}$; unfinished searches are inconclusive.

*Remark* 4.5. The bounds in Table 2 hold for exact-decimal and binary64 data. Case 1440 at $h = 0.025$ reuses its $h = 0.05$ polytope.

For case 148, we close $e_1, e_3$. Since $B_2 e_2 = (1 + h(A_2)_{22})e_2$ with $(A_2)_{22} \approx -1.3680097$, a vertex on the axis of $e_2$ would cap the level at $t \leq 1 + h(A_2)_{22}$ (Appendix B). The 1836-vertex polytope at $h = 0.1$ avoids that axis and certifies $g(0.1) \geq -1.247834$. Transfer by Proposition 4.3(ii), also checked directly at $h = 0.05, 0.025$, gives this bound for every $h \leq 0.1$. Yet $\mathbf{1}^\top e^{tA_2} e_2 = e^{(A_2)_{22}t}$ implies $\check{\lambda}_{\min} \leq (A_2)_{22} < -1.368009$. Thus $\lim g > \check{\lambda}_{\min}$ and both $\lim g$ and $\check{\lambda}$ lie in $[-1.247834, -1.241336]$ (Table 4); the adaptive lower-estimate stall does not reflect failure of Euler convergence.

In contrast with case 148, every finite polytope in the next example is capped by the minimum face rate, although the Euler bound converges.

**Example 4.6.** Blanchini, Colaneri and Valcher [2] consider the three modes

$$A_1 = \left(\begin{smallmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & \varepsilon \end{smallmatrix}\right), \qquad A_2 = \left(\begin{smallmatrix} \varepsilon & 0 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{smallmatrix}\right), \qquad A_3 = \left(\begin{smallmatrix} 0 & 0 & 0 \\ 1 & \varepsilon & 0 \\ 0 & 0 & -1 \end{smallmatrix}\right), \qquad \varepsilon = \tfrac{1}{10}.$$

No convex combination is Hurwitz ($\min_\alpha \eta(S_\alpha) = 1/120$), yet a cyclic schedule stabilizes the system [2]. All three modes are reducible.

- For $x > 0$, $a(x) = (x_1 x_2 x_3)^{1/3}$ satisfies $\frac{\mathrm{d}}{\mathrm{d}t} \log a(x(t)) = \frac{1}{3}(-1 + \varepsilon + x_2/x_3)$ along $A_1$, and the cyclic analogues along $A_2$ and $A_3$. Hence $a(x(t)) \geq e^{(\varepsilon-1)t/3} a(x(0))$ under every switching signal. Since $3a(x) \leq \mathbf{1}^\top x$, Proposition 2.2(ii) gives

  $$\check{\lambda} \geq (\varepsilon - 1)/3.$$

  For the reverse inequality, use the cyclic schedule $A_3$, $A_2$, $A_1$, with equal dwell time $t > 0$. Put $a_t = e^{-t}$, $b_t = e^{\varepsilon t}$ and $f_t = (b_t - 1)/\varepsilon$. Direct multiplication gives

  $$P_t = e^{tA_1} e^{tA_2} e^{tA_3} = \begin{pmatrix} a_t b_t & 0 & f_t a_t^2 \\ f_t a_t & a_t b_t & 0 \\ f_t^2 a_t & f_t a_t b_t & a_t b_t \end{pmatrix}, \qquad \det(zI - P_t) = (z - a_t b_t)^3 - f_t^3 a_t^3 z.$$

  Since $P_t \geq a_t b_t I$, its Perron root is $\rho(P_t) = a_t b_t (1 + k_t)$ with $k_t \geq 0$. The characteristic equation becomes

  $$\frac{k_t^3}{1 + k_t} = \frac{f_t^3 a_t}{b_t^2} \leq \frac{a_t b_t}{\varepsilon^3} \longrightarrow 0.$$

  The function $k \mapsto k^3/(1 + k)$ is strictly increasing on $[0, \infty)$ and vanishes only at zero, so $k_t \to 0$. Repeating this schedule and using Gelfand's formula yields

  $$\check{\lambda} \leq \frac{\log \rho(P_t)}{3t} = \frac{\varepsilon - 1}{3} + \frac{\log(1 + k_t)}{3t} \longrightarrow \frac{\varepsilon - 1}{3}.$$

  Thus $\check{\lambda} = -0.3$, approached by these schedules as their dwell times tend to infinity. Note that at dwell time 30, the exponential enclosure and independent replay of Section 4 give the finite-schedule bound $\check{\lambda} \leq -0.299986$.

- Let $0 < h < 1$, $B_i = I + hA_i$ and $r_h = (1 - h)(1 + \varepsilon h) \in (0, 1)$. The geometric mean satisfies $a(B_i x)^3 \geq r_h a(x)^3$, and iteration from $x = \mathbf{1}$ gives $\check{\rho}(\mathcal{B}_h) \geq r_h^{1/3}$. For the reverse inequality, the block product $P_n = B_1^n B_2^n B_3^n$ has the form of $P_t$, with

  $$a_n = (1 - h)^n, \qquad b_n = (1 + \varepsilon h)^n, \qquad f_n = (b_n - 1)/\varepsilon.$$

  Its Perron root is $a_n b_n (1 + k_n)$, where $k_n \geq 0$ and $k_n^3/(1 + k_n) \leq r_h^n/\varepsilon^3 \to 0$. Consequently

  $$\check{\rho}(\mathcal{B}_h) \leq \rho(P_n)^{1/(3n)} = r_h^{1/3} (1 + k_n)^{1/(3n)} \longrightarrow r_h^{1/3}.$$

  We have therefore proved the exact identities

  $$\check{\rho}(\mathcal{B}_h) = r_h^{1/3}, \qquad g(h) = \frac{r_h^{1/3} - 1}{h} \longrightarrow \frac{\varepsilon - 1}{3} = \check{\lambda}.$$

  The values of $g$ round to $-0.313030$ at $h = 0.1$ and $-0.306335$ at $h = 0.05$. The axes are invariant faces with exponent $-1$, each under two modes, and the two-dimensional faces have exponent $1/120$; hence $\check{\lambda}_{\min} = -1$ by Theorem 2.7.

- Every finite $V$ that satisfies (4) has $t \leq 1 - h$, so Proposition 4.3 certifies at most $\check{\lambda} \geq -1$, the exponent of the axes, at every step. Indeed, suppose $t > 1 - h$. Then:

– Let $m_3 = \min_{v \in V} v_3$. Since $(B_3 x)_3 = (1-h)x_3$, applying (4) at a minimizer gives $(1-h)m_3 \geq t m_3$, so $m_3 = 0$ and $V_0 = \{v \in V : v_3 = 0\}$ is nonempty.

– Let $m_2 = \min_{u \in V_0} u_2$. Since $(B_2 x)_3 = x_3$, the weights for a minimizer $u$ are supported on $V_0$. Since $(B_2 x)_2 = (1-h)x_2$, we get $(1-h)m_2 \geq t m_2$, so $m_2 = 0$. As all vertices are nonzero, $V$ has a point on the positive axis of $e_1$.

– For the smallest of them, $me_1$, (4) for $(1, me_1)$ reads $(1-h)me_1 \geq t \sum_w \lambda_w w$. So the weights live on that axis, and $(1-h)m \geq tm$, a contradiction.

Applied to the families $\{A_i - \alpha I\}$ as in Numerical observation 4.9, the same argument caps the tangent bound of Algorithm (L) at $-1$; the positive antinorms used by the Gripenberg algorithm [9] are capped at $-1$ by the bound $\check{\rho}_{\min}$ in Appendix B. In our runs at $h = 0.1$, closures seeded by $\mathbf{1}$, by points of the two-dimensional faces, or by the coordinate vectors finish at the level $g = -1.01$, and none finishes at $-0.99$ within 120 s.

So finite polytope bounds do not approach the true exponent here, although the Euler bound converges. The optimal trajectories dwell near the axes, which are fast faces; the extremal antinorm, the geometric mean, vanishes on all of $\partial K$ and is not polyhedral.

### 4.3 Numerical results

The experiments are designed to answer three questions: whether certified brackets behave as the rate theorems predict (Numerical observation 4.7), how the polytope search scales with the dimension (Numerical observation 4.8), and how the Euler approach compares with Algorithm (L) of [12] (Numerical observation 4.9).

Appendices F and G record generators, budgets, hardware and search protocols. Adaptive Gripenberg lower estimates guide searches but require a posteriori verification: every test-pair estimate is verified, whereas seven of 54 random-pair estimates remain unverified within the allocated budget (see Appendix B).

**Numerical observation 4.7.** On test pairs, the bracket widths in Table 2 are approximately $1.48h$ for the planar pair (whose known exponent is enclosed) and $0.55h$ for case 1, matching $Qh$ with the coefficients $Q = 1.478$ and 0.548 of Theorem 3.12; case 1440's width halves from $h = 0.1$ to 0.05. Subsequent improvements for cases 1440 and 148 reuse polytopes from larger steps. Their certified upper bounds beat numerical mixture minima by up to $2.2 \times 10^{-3}$ and $3.1 \times 10^{-3}$, respectively; proving $\check{\lambda} < c$ still requires a global lower bound for $c$. For the fourteen pairs of Appendix C, finite-word slopes also approach the conjectured coefficients.

The rates cannot, however, be used directly to sharpen these bounds. For dimension-5 pair 4 of Table 3, the certified endpoints $L_h$ at target $10^{-5}$, rounded outward, satisfy $2L_{0.05} - L_{0.1} = -0.426998 > -0.430082$, the certified schedule upper bound, so the extrapolated lower endpoint is not valid.

**Numerical observation 4.8.** The performance degrades quickly with the dimension. Indeed, at $h = 0.1$, certified lower bounds are within $4.4 \times 10^{-3}$ of the word estimates for 18/20 dimension-5 pairs of the scaling sample, but for only 1/20 dimension-10 pairs, and dimension-20 widths are between 0.33 and 1.6.

The scaling sample has ten dense and ten sparse pairs in each of dimensions $5, 10$, plus five dimension-20 pairs, with irreducible modes and interior numerical mixture minima. Chattering remains unproved; dimension-10 sparse 8 admits an improving schedule. Results are in Tables 9 and 10 and the repository records. Transfer improves every direct $h = 0.05$ bound for the original six dimension-5 pairs, and mixture bounds improve on implicit Euler for the original twelve pairs (Table 4).

The switching sample (Table 3 and Table 11) contains 30 pairs in each of dimensions $5, 10$ and five in dimension 20, selected by schedules improving the numerical mixture minimum by

Table 2: Certified Euler brackets (Remark 4.5); $c$ is the numerical mixture minimum, rounded up. $^\dagger$Lower bound obtained by reusing the polytope from $h = 0.1$ (148) or 0.05 (1440). Upper endpoints use implicit words; width is upper minus lower. The planar pair is that of Numerical observation 3.13.

| pair ($c$) | $h$ | lower | upper | width |
|---|---|---|---|---|
| planar | 0.1 | $-0.272230$ | $-0.123619$ | $1.5 \times 10^{-1}$ |
| ($-0.194976$) | 0.05 | $-0.232706$ | $-0.158706$ | $7.4 \times 10^{-2}$ |
| | 0.025 | $-0.213641$ | $-0.176676$ | $3.7 \times 10^{-2}$ |
| | 0.0125 | $-0.204261$ | $-0.185783$ | $1.8 \times 10^{-2}$ |
| case 1 | 0.1 | $-1.044234$ | $-0.988047$ | $5.6 \times 10^{-2}$ |
| ($-1.012395$) | 0.05 | $-1.027040$ | $-0.999505$ | $2.8 \times 10^{-2}$ |
| | 0.025 | $-1.019518$ | $-1.005757$ | $1.4 \times 10^{-2}$ |
| 1440 | 0.1 | 0.344598 | 0.346673 | $2.1 \times 10^{-3}$ |
| (0.348177) | 0.05 | 0.345146 | 0.346182 | $1.0 \times 10^{-3}$ |
| | 0.025 | $0.345146^\dagger$ | 0.345931 | $7.9 \times 10^{-4}$ |
| 148 | 0.1 | $-1.247834$ | $-1.236958$ | $1.1 \times 10^{-2}$ |
| ($-1.236979$) | 0.05 | $-1.247834^\dagger$ | $-1.238961$ | $8.9 \times 10^{-3}$ |
| | 0.025 | $-1.247834^\dagger$ | $-1.240094$ | $7.7 \times 10^{-3}$ |

$10^{-3}$; this does not prove $\check{\lambda} < c$. At $h = 0.1$, a target $10^{-3}$ below the word estimate is reached for 30/30 dimension-5 pairs and 14/30 dimension-10 pairs, while dimension 20 closes only at target 0.1. Target tolerance is measured below a word estimate, not relative to $\check{\lambda}$; outward rounding adds at most $10^{-6}$ per endpoint, and the bounds hold for both data readings. Finer steps increase search cost, and bracket widths also include schedule suboptimality. Table 3 and Table 11 report separate searches at each step; transfer preserves the best lower bound as $h$ decreases (Proposition 4.3(ii)). Detailed success counts and widths are in Appendix G.

### 4.4 Comparison with Algorithm (L)

Algorithm (L) of [12, Sec. 4.6] certifies lower bounds by polytopes for the exponential family $\{e^{\tau A_i}\}$. We ran a comparison of our simplest implementation of it with the closure code and budgets of the Euler searches.

**Numerical observation 4.9.** Neither the Euler approach nor Algorithm (L) dominates on the test and scaling pairs. Under the reported implementations and budgets, Euler performs better on the switching benchmarks in Figure 2; the best periods found can exceed the reach of short exponential words. At $\tau = 0.1, 0.05$, Algorithm (L) gives the larger lower bound on 6/15 pairs, versus Euler's 9/15 (Table 4; reducible case 148 is excluded). Its tangent test improves neither the planar nor case 1 Euler polytope. Rigorous mixture or schedule bounds improve both methods' upper bounds.

In Figure 2, Euler gaps decrease roughly linearly, with time rising from under one second to 280–1150 s. Algorithm (L) is comparable for case 1 but worse for the switching pairs: pair 4's gap exceeds 0.27, since length-14 words at $\tau = 0.025$ cover periods only up to 0.35, versus the best period near 3.3. The figure uses fixed certified upper bounds and includes searches, failures and checks in the timings.

The lower bound of Algorithm (L) is the largest $\alpha$ such that $(A_i - \alpha I)w$ points into $\mathcal{P} = \operatorname{conv} V + K$ at every vertex $w$, for every $i$. This is the bound $\check{\alpha}(\mathcal{P})$ of [12, Prop. 8]. It is Proposition 4.3 in the limit $h \to 0$: if the condition holds, then $(I + s(A_i - \alpha I))\mathcal{P} \subseteq \mathcal{P}$ for small $s > 0$, and Proposition 4.3(i), applied to the family $\{A_i - \alpha I\}$ with $t = 1$, gives $\check{\lambda} \geq \alpha$.

We compute $\alpha$ by linear programming and then exactly, from rationalized multipliers, for both data readings. These lower bounds are certified but not independently replayed.

Table 3: Pairs selected by improving schedules (Numerical observation 4.8). Estimates: $c$ (mixture) and $g(h) \leq$ (explicit word), rounded up. Bounds: the schedule bound "upper" and $g(h) \geq$, the better lower bound from both orientations, rounded down; both hold for both data readings. $^{\top}$Transpose; $^{+}$extended search. Width is upper minus the larger lower bound, rounded up.

| | | | | $h = 0.1$ | | $h = 0.05$ | | |
|---|---|---|---|---|---|---|---|---|
| $d$ | pair | $c$ | upper | $g(h) \leq$ | $g(h) \geq$ | $g(h) \leq$ | $g(h) \geq$ | width |
| 5 | 1 | 0.539231 | 0.467683 | 0.465337 | 0.465326 | 0.466492 | 0.366491 | $2.4 \times 10^{-3}$ |
| 5 | 2 | 0.459280 | 0.457887 | 0.442429 | 0.442418 | 0.450720 | 0.440719 | $1.6 \times 10^{-2}$ |
| 5 | 3 | −0.333377 | −0.365833 | −0.457212 | −0.457223 | −0.409526 | −0.410527$^{\top}$ | $4.5 \times 10^{-2}$ |
| 5 | 4 | −0.401497 | −0.430082 | −0.460383 | −0.460394 | −0.443685 | −0.443696 | $1.4 \times 10^{-2}$ |
| 5 | 5 | 0.587104 | 0.548729 | 0.531162 | 0.531151 | 0.540134 | 0.539133 | $9.6 \times 10^{-3}$ |
| 5 | 6 | 0.532754 | 0.519922 | 0.500647 | 0.500636 | 0.511408 | 0.511397 | $8.6 \times 10^{-3}$ |
| 5 | 7 | 1.431824 | 1.428992 | 1.377996 | 1.377985 | 1.406985 | 1.405984$^{\top}$ | $2.4 \times 10^{-2}$ |
| 5 | 8 | −0.152645 | −0.171842 | −0.210279 | −0.211280 | −0.189358 | −0.189369 | $1.8 \times 10^{-2}$ |
| 5 | 9 | 0.481345 | 0.474726 | 0.404665 | 0.404654 | 0.440660 | 0.430659 | $4.5 \times 10^{-2}$ |
| 5 | 10 | 0.857180 | 0.843598 | 0.839946 | 0.838945 | 0.841758 | 0.840757 | $2.9 \times 10^{-3}$ |
| 10 | 1 | 0.567910 | 0.559043 | 0.532613 | 0.522612$^{+}$ | 0.546054 | 0.446053 | $3.7 \times 10^{-2}$ |
| 10 | 2 | 0.716928 | 0.674041 | 0.629877 | 0.629866 | 0.653399 | 0.553398 | $4.5 \times 10^{-2}$ |
| 10 | 3 | 0.802565 | 0.788865 | 0.777934 | 0.727933$^{+}$ | 0.783449 | 0.683448 | $6.1 \times 10^{-2}$ |
| 10 | 4 | 0.253414 | 0.245147 | 0.216605 | 0.196604$^{+}$ | 0.231252 | 0.081251$^{\top}$ | $4.9 \times 10^{-2}$ |
| 10 | 5 | −0.168419 | −0.197966 | −0.228868 | −0.228879 | −0.212874 | −0.312875 | $3.1 \times 10^{-2}$ |
| 10 | 6 | 0.673707 | 0.638392 | 0.635617 | 0.625616$^{\top}$ | 0.636998 | 0.536997$^{\top}$ | $1.3 \times 10^{-2}$ |
| 10 | 7 | 1.200095 | 1.185699 | 1.136618 | 1.136607$^{\top}$ | 1.161028 | 1.061027$^{\top}$ | $5.0 \times 10^{-2}$ |
| 10 | 8 | 0.198883 | 0.142660 | 0.077000 | 0.056999$^{+}$ | 0.110582 | 0.010581$^{+}$ | $8.6 \times 10^{-2}$ |
| 10 | 9 | 0.953083 | 0.944065 | 0.932144 | 0.882143$^{\top+}$ | 0.938291 | 0.738290$^{\top}$ | $6.2 \times 10^{-2}$ |
| 10 | 10 | 0.670254 | 0.387498 | 0.369426 | 0.359425$^{+}$ | 0.378543 | 0.278542$^{+}$ | $2.9 \times 10^{-2}$ |

Closure code, budgets and hardware are shared, but our extra long-word search and naive implementation prevent attributing differences solely to discretization. Sections F.5 and F.7 specify the exponential-word search, the polytope closure and the timing protocol.

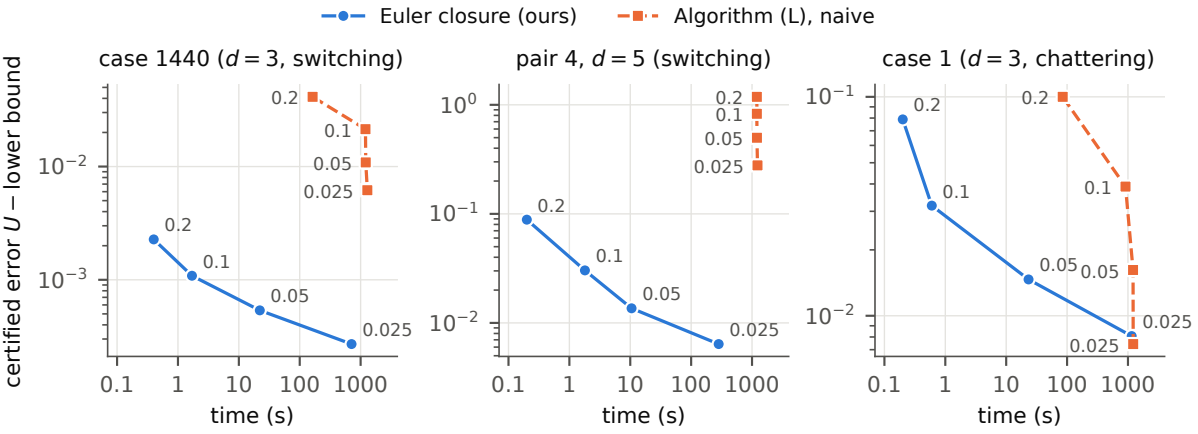



Figure 2: Certified bracket width versus time (Numerical observation 4.9), log-log: Euler closure and our naive Algorithm (L). Labels give $h$ or $\tau$; times below 0.1 s are drawn at 0.1 s.

Table 4: Best bounds over steps $0.1, 0.05$ (Numerical observation 4.9), with the larger lower bound in bold. Algorithm (L) lower bounds are not independently replayed, and its upper values are estimates. $^{g}$Closure seeded with adaptive Gripenberg vertices at $h = 0.1$; our search alone gives $0.254546, 1.312394$. "$c$ or schedule" is a mixture or $^{s}$schedule bound. Width uses the larger lower bound and last upper column; upper values and widths are rounded up. Reducible case 148 was not run with Algorithm (L).

| | lower bound (certified) | | upper bound | | | |
|---|---|---|---|---|---|---|
| pair | Alg. (L) | ours | Alg. (L) | implicit Euler | $c$ or schedule | width |
| planar | $\mathbf{-0.210758}$ | $-0.232706$ | $-0.194213$ | $-0.158706$ | $-0.194976$ | $1.6 \times 10^{-2}$ |
| case 1 | $-1.028582$ | $\mathbf{-1.027040}$ | $-1.011626$ | $-0.999505$ | $-1.012395$ | $1.5 \times 10^{-2}$ |
| case 1440 | $0.334850$ | $\mathbf{0.345146}$ | $0.346401$ | $0.346182$ | $0.345675^{s}$ | $5.3 \times 10^{-4}$ |
| case 148 | — | $-1.247834$ | — | $-1.238961$ | $-1.241336^{s}$ | $6.5 \times 10^{-3}$ |
| $d = 5$, dense 1 | $-0.276163$ | $\mathbf{-0.222751}$ | $-0.213075$ | $-0.209005$ | $-0.213383$ | $9.4 \times 10^{-3}$ |
| $d = 5$, dense 2 | $-0.081527$ | $\mathbf{-0.059850}$ | $-0.028459$ | $-0.014798$ | $-0.029331$ | $3.1 \times 10^{-2}$ |
| $d = 5$, dense 3 | $0.210187$ | $\mathbf{0.222107}$ | $0.233137$ | $0.237887$ | $0.232972$ | $1.1 \times 10^{-2}$ |
| $d = 5$, sparse 1 | $-0.204184$ | $\mathbf{-0.107049}$ | $-0.032087$ | $-0.001609$ | $-0.034259$ | $7.3 \times 10^{-2}$ |
| $d = 5$, sparse 2 | $\mathbf{-0.384238}$ | $-0.641285$ | $-0.102190$ | $-0.097694$ | $-0.102660$ | $2.9 \times 10^{-1}$ |
| $d = 5$, sparse 3 | $\mathbf{-0.049779}$ | $-0.071028$ | $0.055587$ | $0.112334$ | $0.053208$ | $1.1 \times 10^{-1}$ |
| $d = 10$, dense 1 | $\mathbf{0.870186}$ | $0.459247$ | $0.990484$ | $0.991569$ | $0.990410$ | $1.3 \times 10^{-1}$ |
| $d = 10$, dense 2 | $\mathbf{0.779913}$ | $0.525978$ | $0.857197$ | $0.865203$ | $0.856999$ | $7.8 \times 10^{-2}$ |
| $d = 10$, dense 3 | $\mathbf{0.009496}$ | $-0.229448$ | $0.066898$ | $0.068764$ | $0.066852$ | $5.8 \times 10^{-2}$ |
| $d = 10$, sparse 1 | $-0.064519$ | $\mathbf{0.269192}^{g}$ | $0.318964$ | $0.334496$ | $0.317685$ | $4.9 \times 10^{-2}$ |
| $d = 10$, sparse 2 | $0.364290$ | $\mathbf{1.356787}^{g}$ | $1.570804$ | $1.583464$ | $1.570368$ | $2.2 \times 10^{-1}$ |
| $d = 10$, sparse 3 | $-0.451797$ | $\mathbf{-0.349697}$ | $-0.179980$ | $-0.143149$ | $-0.181089$ | $1.7 \times 10^{-1}$ |

# 5 Open problems

We conclude by summarizing some open questions that arise naturally from this paper.

(1) **General explicit convergence.** Does $g(h) \to \check{\lambda}$ as $h \downarrow 0$ for every finite family of Metzler matrices? The sufficient cone criterion of Corollary 3.8 settles the five test pairs; its failure leaves convergence undecided.

(2) **Sharp first-order asymptotics.** Under Theorem 3.12, with $\check{\lambda} = c$ and $\eta''(a^*) > 0$, where $\eta(a) = \eta(S_a)$, do $g(h) = c - Qh/2 + O(h^2)$ and $G(h) = c + Qh/2 + O(h^2)$ hold? Matching lower estimates are missing. For three or more modes, identifying optimal coefficients must also address word ordering (Appendix D).

(3) **Gripenberg extensions.** Can the fixed-antinorm result of Corollary 4.1 be extended to reducible families? What conditions ensure validity and convergence of the adaptive lower estimates (Remark 4.2)?

(4) **Computational cost.** Which structural assumptions yield bounds on vertex counts and computational cost for a prescribed dimension and certified bracket width? The obstruction in Example 4.6 motivates nonpolyhedral antinorms. Combining the constructions of Numerical observation 4.9 is a separate practical direction; it cannot remove a shared face cap.

(5) **Other discretizations and cones.** Can cone-preserving higher-order discretizations yield rigorous two-sided bounds with smaller discretization gaps? On proper nonpolyhedral cones, flow invariance need not imply invariance under $I + hA$.

## Acknowledgments

The author is a member of INdAM-GNCS (Gruppo Nazionale di Calcolo Scientifico).

## Statements and declarations

### Data and code availability

The datasets generated and analysed during this study, together with the computational code, are available in the `certified-lower-exponents` repository, at verified commit `a10a464e1f7a`. The repository contains the input matrices, complete numerical records, stored certificates, and independent Python and MATLAB verification scripts.

The repository also contains the original Gripenberg run records and final vertex sets. The repository is publicly accessible. The main code uses the MIT License and the optional Gripenberg implementation retains GPL-3.0.

Detailed Gripenberg runs appear in Appendix B; Appendices F and G give the experimental settings, verification conventions, and scaling and switching samples.

Commands and driver options are in `reproduce/README.md`, with instructions for the Gripenberg runs in `reproduce/gm/README.md`. Detailed results are in `reproduce/results/`, including `gm_original/` and `larger_samples/`. The verification run of 30 October 2026 passed all 14 Python regression tests and replayed all 568 supplied certificates. Replays check saved witnesses; rerunning searches can change timings and final estimates.

Algorithm (L)'s lower bounds are checked exactly when generated but lack independent replay witnesses; its upper bounds and the word and mixture estimates remain floating point.

### Author contributions

F.P.M. conceived the study, developed the mathematical results and proofs, developed and validated the software, carried out the numerical experiments, prepared the figures and tables, and wrote and revised the manuscript. F.P.M. approved the final manuscript and takes responsibility for all aspects of the work.

### Funding

No funding was received for this research.

### Competing interests

The author has no competing interests to declare.

### AI assistance

Codex (OpenAI) assisted with numerical experiments, code development, and presentation/grammar editing. The author takes full responsibility for the article; in particular, verified bounds rest on the exact checks of Section 4.

## A Data of the test pairs

Cases 77, 148, 677, 1195 and 1440 retain the identifiers used in the repository. Case 1 is an additional test pair, specified to six decimal places. For the other cases, each entry is printed with the fewest digits needed to recover its stored double-precision (binary64) value when read back.

The bounds in Section 4 are valid whether the printed decimals are treated as exact numbers or converted to binary64. The computations in Computation 3.9 use the latter interpretation.

The planar pairs are given in Example 3.10 and Numerical observation 3.13.

**Case 1**:

$$A_1 = \begin{pmatrix} -1.730493 & 0.996436 & 0.459716 \\ 0.69104 & -2.643051 & 0.03405 \\ 0.84589 & 0.587882 & -2.309322 \end{pmatrix},$$
$$A_2 = \begin{pmatrix} -2.892226 & 0.839125 & 0.466303 \\ 0.127203 & 0.220544 & 0.195653 \\ 0.06192 & 0.598392 & -2.23932 \end{pmatrix}.$$

**Case 77**:

$$A_1 = \begin{pmatrix} -0.10421255427821108 & 0 & 0.5995296020142661 \\ 0 & -3.983676417656142 & 10.44547437798186 \\ 0.6089541200819454 & 0.41616545550352757 & -3.2474691918642016 \end{pmatrix},$$
$$A_2 = \begin{pmatrix} -0.6965629728040771 & 0 & 0.23901622681524412 \\ 0 & 0.02489242262644542 & 0 \\ 0.18904586288556002 & 0.8602987239031196 & -2.0161017416771956 \end{pmatrix}.$$

**Case 148**:

$$A_1 = \begin{pmatrix} -2.854194178918802 & 0.24749928971794594 & 0 \\ 0.56180752192356 & -1.3059431283783423 & 0.20794658684868464 \\ 0 & 0 & -3.669288482126051 \end{pmatrix},$$
$$A_2 = \begin{pmatrix} -3.653224173747234 & 0 & 0.49425460335236504 \\ 2.342863144895686 & -1.3680096637401422 & 0 \\ 0.7499705168800616 & 0 & 0.05001607999557478 \end{pmatrix}.$$

**Case 677**:

$$A_1 = \begin{pmatrix} -3.5751753658589895 & 0 & 0.7873089953050745 \\ 3.872088293234137 & -2.4704665923380382 & 3.292200232648817 \\ 1.6772091050048494 & 0.4874274815499069 & -1.7576243608193685 \end{pmatrix},$$
$$A_2 = \begin{pmatrix} -3.608486201678475 & 0 & 4.290473593750214 \\ 0 & -0.23709653436107514 & 4.089406788141199 \\ 0.7796352280553934 & 0 & -3.3591039935935543 \end{pmatrix}.$$

**Case 1195**:

$$A_1 = \begin{pmatrix} -3.2011480664004672 & 0 & 0 \\ 0 & -1.601165653605288 & 1.1392361293277327 \\ 0.11566889906380853 & 2.7419424885042747 & -2.670478064763273 \end{pmatrix},$$
$$A_2 = \begin{pmatrix} 0.09930018711201494 & 0 & 0.271862921876863 \\ 0.831655604528253 & -2.3694492859575673 & 2.394339449701255 \\ 5.327089103180349 & 1.129064780325959 & -3.977234504357898 \end{pmatrix}.$$

**Case 1440**:

$$A_1 = \begin{pmatrix} -2.163042579047693 & 0.33559573959929034 & 0.17548453245292883 \\ 0.7960011589145731 & -1.0721730232799498 & 0.7689150093877756 \\ 0.3079265062822859 & 0.7102921800837112 & -0.11334509645810664 \end{pmatrix},$$
$$A_2 = \begin{pmatrix} -0.365512369671777 & 0.895993665091137 & 0.05148281594810899 5 \\ 0.24535640370572198 & -0.3430416586652343 & 0.3203492321022128 \\ 0.05334247416806037 & 0.7483745160181475 & -0.29764076048251553 \end{pmatrix}.$$

# B Runs of the Gripenberg algorithms

Corollary 4.1 guarantees fixed-antinorm convergence. Tables 5 to 7 report adaptive runs, whose lower estimates require a posteriori verification (Remark 4.2).

**Normalization and accuracy** Runs use transposed families and $\delta = h^2/10$, a discrete-gap tolerance rather than a continuous-time width; precise settings are in Section B.1.

For a fixed antinorm, scaling $(\mathcal{F}, \delta)$ to $(\gamma\mathcal{F}, \gamma\delta)$ scales both bounds by $\gamma > 0$. Adaptive acceptance requires $a(z) \le 1$ and presumes $\check{\rho}(\mathcal{F}) \approx 1$ [9, Sec. 4.2]; we normalize as $e^{-h\hat{\lambda}}\mathcal{B}_h$ and $(1-h\hat{\lambda})\mathcal{R}_h$, then rescale. Without normalization, case 1440's lower estimate remained nonpositive at $h = 0.1, 0.05$, although $\check{\lambda} \approx 0.3457$. Transposition preserves primitivity, $\check{\rho}$ and, under Corollary 4.1, the cone condition. On $\gamma\mathcal{F}$, tolerance $\delta$ becomes $\delta/\gamma$ in that corollary, contributing $\delta/(\gamma h)$ to the explicit-rate error; without normalization this corresponds to $\delta/h = h/10$.

The code's inherited joint-spectral-radius error correction inflates antinorms by order $10^{-15}$ here. This cannot validate lower bounds as exact polytope checks do.

**Numerical observation B.1** (Gripenberg runs)**.** Adaptive runs often exhaust their budgets at finer steps. All four fixed-antinorm explicit runs do so with weaker verified lower bounds, whereas their implicit estimates are within $2 \times 10^{-4}$ of Table 2. Tables 5 to 7 give the counts and seeded-closure improvements.

**The obstruction in case 148** Both modes are reducible. Since the third row of $A_1$ is $(0, 0, \nu)$, with $\nu = (A_1)_{33} \approx -3.6692885 < \check{\lambda}$ and $\eta(A_1) \approx -1.220815$, $A_1^\top e_3 = \nu e_3$ is a secondary eigenvector in $K$, violating the cone condition [9]. For $P = (I + hA_1)^\top$, $\mu = 1 + h\nu$ and every positive antinorm $a$, including adaptive ones,

$$a(P^k) \le \frac{a(P^k e_3)}{a(e_3)} = \mu^k,$$

so $q(P^k) \le \mu$ in [9]. Since $\mu + \delta$ lies below every upper bound reached, $P^k$ is never discarded in exact arithmetic: $t_k \le \mu$ and $(t_k - 1)/h \le \nu$. The observed plateau is $-3.6692885$, as in the critical example of [9, Sec. 5.2]; upper bounds remain valid, but $s_k - t_k > 5 \times 10^{-2}$ at every reported step.

More generally, with a fixed positive antinorm the Gripenberg lower bounds never exceed

$$\check{\rho}_{\min}(\mathcal{F}) = \lim_{n\to\infty} \min_{|w|=n} \min_j \big(\mathbf{1}^\top M_w e_j\big)^{1/n},$$

the discrete counterpart of $\check{\lambda}_{\min}$ and of Lemma 2.3(ii). Indeed, the proof of the lower-bound lemma of [9, Sec. 3.1] gives $t_k \le \liminf_N \alpha_N(\mathcal{F})$, where $\alpha_N(\mathcal{F}) = \min_{|w|=N} a(M_w)^{1/N}$, without the cone condition; and $c_1 \mathbf{1}^\top x \le a(x) \le c_2 \mathbf{1}^\top x$ on $K$ gives

$$\alpha_N(\mathcal{F}) \le (c_2/c_1)^{1/N} \min_{|w|=N} \min_j \big(\mathbf{1}^\top M_w e_j\big)^{1/N}.$$

Without transposition, $A_2 e_2 = (A_2)_{22} e_2$ gives the better cap $(A_2)_{22} = \check{\lambda}_{\min} \approx -1.368010$ (Computation 3.9). Transposition preserves the lower spectral radius, so one can retain the better orientation. Even then, positive antinorms leave width at least $\check{\lambda} - (A_2)_{22} \ge 0.12$, since Remark 4.5 gives $\check{\lambda} \ge -1.247834$. Its polytopes avoid the axis of $e_2$; their antinorms vanish there and escape this cap.

## B.1 Implementation, settings and detailed results

The following results complement Numerical observation B.1; verification procedures are given in Appendix F.

All Gripenberg runs use the updated `subradius_computation` repository of Guglielmi and Maiale, fixed at commit `181d6b5`. Runs use transposed families, coordinate seeds, accuracy $\delta = h^2/10$ and a budget of $2 \times 10^4$ products, which can be exceeded while completing a level. Time caps are 7200 s for test pairs and 1800 s for random pairs. The fixed-antinorm runs discussed in Numerical observation B.1 use $a(x) = \mathbf{1}^\top x$ and 3600 s. Time caps are checked between products; twelve runs ran concurrently. Accuracy $\delta$ means a reported discrete gap for the supplied, possibly normalized family, not a certified continuous-time bracket width. The explicit and implicit families are normalized as $e^{-h\hat{\lambda}}\mathcal{B}_h$ and $(1 - h\hat{\lambda})\mathcal{R}_h$ and then rescaled. We use $\hat{\lambda} = 0.345674608$ for case 1440, no normalization for the other test pairs, the mixture value for the scaling sample and the certified schedule bound for the switching sample. The preceding analysis explains the normalization, rounding caveat and face obstruction in case 148.

*Test pairs* (Table 5). The planar pair and cases 1 and 1440 reach accuracy $\delta$ at $h = 0.1$ and 0.05. At finer steps the explicit runs exhaust their product or time budgets and fall well below the certified bounds of Table 2. Every reported lower estimate is certified a posteriori by Proposition 4.3, with 3 to 225 vertices.

*Random pairs* (Table 6). At $h = 0.1$, four of six dimension-5 explicit runs reach accuracy $\delta$; none does at $h = 0.05$. Only one of the 24 dimension-10 runs, explicit and implicit, reaches it. Nine of the other eleven explicit runs discard no products, exploring complete trees of depth 9 or 10. Our lower bounds exceed the estimates in 18 of 24 rows. Seeding with the final vertices

of the adaptive Gripenberg runs certifies two further estimates, 0.269192 and 1.356787, used in Table 4; the remaining four higher estimates are uncertified.

*Switching pairs* (Table 7). Seven of ten dimension-5 explicit runs reach accuracy $\delta$ at $h = 0.1$; all others time out below our certified bounds. All implicit runs time out above the schedule bounds.

*Fixed antinorm.* All four explicit runs exhaust the product budget before accuracy $\delta$; their certified estimates lie below the adaptive and our bounds. Implicit estimates are within $2 \times 10^{-4}$ of Table 2. Slow fixed-antinorm convergence is also discussed in [9, Sec. 5.1].

**Certification of the outputs** A final adaptive vertex set can be checked directly by Proposition 4.3; pruning and rescaling need not preserve invariance. Alternatively, it can seed a new closure. Seeded closures certify the estimate, rounded down to six decimals, in 45/47 explicit test, switching and fixed-antinorm runs. Failures within 600 s are case 148 and dimension-10 pair 3 at $h = 0.1$; none improves Table 2 and Table 3. Failure is inconclusive since separate closures certify every test-pair estimate, as reported above.

Table 5: Adaptive Gripenberg runs (Numerical observation B.1). Lower: explicit estimate; upper: best implicit product, both converted to continuous-time rates. Depth gives the two maximum word lengths; time is their sum. *Product budget and †time limit reached before accuracy $\delta$.

| pair | $h$ | lower | upper | depth | time (s) |
|---|---|---|---|---|---|
| planar | 0.1 | −0.281832 | −0.121884 | 11/10 | 12 |
| | 0.05 | −0.237069 | −0.157436 | 13/25 | 43 |
| | 0.025 | −0.238368* | −0.176426 | 20/19 | 4445 |
| | 0.0125 | −1.019681* | −0.185482* | 15/15 | 6189 |
| case 1 | 0.1 | −1.054233 | −0.987914 | 9/10 | 14 |
| | 0.05 | −1.031873 | −0.999444 | 16/33 | 215 |
| | 0.025 | −1.100284† | −1.005758† | 14/14 | 16289 |
| 1440 | 0.1 | 0.336905 | 0.347799 | 11/12 | 36 |
| | 0.05 | 0.342769 | 0.347957 | 18/19 | 3235 |
| | 0.025 | 0.116748† | 0.348080† | 13/13 | 14400 |
| 148 | 0.1 | −3.669288* | −1.235611* | 14/15 | 1426 |
| | 0.05 | −3.669288* | −1.235823* | 14/14 | 1376 |
| | 0.025 | −3.669288* | −1.235803* | 14/14 | 258 |

Table 6: Random-pair explicit estimates (Numerical observation B.1), floating point; $c$ is the numerical mixture minimum; *time limit before accuracy $\delta$. "Ours" certifies $g(h)$; †transferred from $h = 0.1$. The last column certifies the adaptive Gripenberg estimate by a closure seeded with its final vertices; — means no closure within 300 s.

| $d$ | pair ($c$) | $h$ | estimate | ours | seeded bound |
|---|---|---|---|---|---|
| 5 | dense 1 (−0.213383) | 0.1 | −0.232425 | −0.222751 | −0.232425 |
| | | 0.05 | −0.292711* | −0.222751† | −0.292712 |
| 5 | dense 2 (−0.029331) | 0.1 | −0.069430 | −0.059850 | −0.069431 |
| | | 0.05 | −0.138457* | −0.059850† | −0.138458 |
| 5 | dense 3 (+0.232972) | 0.1 | 0.194335* | 0.222107 | 0.194334 |
| | | 0.05 | 0.057287* | 0.222107† | — |
| 5 | sparse 1 (−0.034259) | 0.1 | −0.112654 | −0.107049 | −0.112655 |
| | | 0.05 | −0.814649* | −0.107049† | −0.814650 |
| 5 | sparse 2 (−0.102660) | 0.1 | −0.798666* | −0.641285 | −0.798667 |
| | | 0.05 | −1.014139* | −0.641285† | −1.014139 |

Table 6: Continued from the previous page.

| $d$ | pair ($c$) | $h$ | estimate | ours | seeded bound |
|---|---|---|---|---|---|
| 5 | sparse 3 (+0.053208) | 0.1 | −0.080404 | −0.071028 | −0.080404 |
| | | 0.05 | −0.076264* | −0.071028† | −0.076264 |
| 10 | dense 1 (+0.990410) | 0.1 | 0.533412* | 0.459247 | — |
| | | 0.05 | 0.046459* | 0.459247† | — |
| 10 | dense 2 (+0.856999) | 0.1 | 0.783322* | 0.525978 | — |
| | | 0.05 | 0.541437* | 0.525978† | — |
| 10 | dense 3 (+0.066852) | 0.1 | −0.164566* | −0.229448 | — |
| | | 0.05 | −0.612417* | −0.229448† | −0.612417 |
| 10 | sparse 1 (+0.317685) | 0.1 | 0.269193 | 0.254546 | 0.269192 |
| | | 0.05 | −1.092236* | 0.254546† | −1.092236 |
| 10 | sparse 2 (+1.570368) | 0.1 | 1.356787* | 1.312394 | 1.356787 |
| | | 0.05 | 0.762998* | 1.312394† | 0.762998 |
| 10 | sparse 3 (−0.181089) | 0.1 | −0.917161* | −0.349697 | −0.917161 |
| | | 0.05 | −1.723107* | −0.349697† | −1.723108 |

Table 7: Explicit estimates from the adaptive Gripenberg algorithm for Table 3, converted to $g(h)$, and run times (floating point). *Time limit before accuracy $\delta$; dimension 10 was not run at $h = 0.05$.

| | $d = 5$, $h = 0.1$ | | $d = 5$, $h = 0.05$ | | $d = 10$, $h = 0.1$ | |
|---|---|---|---|---|---|---|
| pair | estimate | time (s) | estimate | time (s) | estimate | time (s) |
| 1 | 0.114886* | 1800 | 0.080790* | 2703 | −0.391747* | 3544 |
| 2 | 0.431960 | 103 | 0.021179* | 1801 | −1.797610* | 1800 |
| 3 | −0.466853 | 778 | −1.057132* | 1801 | 0.151578* | 1800 |
| 4 | −0.463750 | 421 | −0.971803* | 4148 | −0.876970* | 5116 |
| 5 | 0.520597 | 288 | 0.400567* | 2252 | −1.331466* | 1801 |
| 6 | 0.419929* | 2010 | −0.459780* | 1800 | −0.766025* | 1800 |
| 7 | 1.366459 | 72 | 0.628847* | 3368 | 1.079967* | 2511 |
| 8 | −0.218978 | 760 | −1.557098* | 1801 | −0.488982* | 1800 |
| 9 | 0.394178 | 1156 | −0.450092* | 1801 | 0.504566* | 3581 |
| 10 | 0.766664* | 1800 | −0.036211* | 1826 | −0.018742* | 1801 |

# C First-order coefficients

For the planar pair, case 1 and the original twelve pairs of Numerical observation 4.8, use $a^*, c, Q$ and Perron vectors normalized by $u^\top v = 1$ as in Theorem 3.12. The finite-word estimates $g_{\mathrm{est}}, G_{\mathrm{est}}$ define

$$s_- := \frac{c - g_{\mathrm{est}}}{h}, \qquad s_+ := \frac{G_{\mathrm{est}} - c}{h}, \qquad s_- + s_+ = \frac{G_{\mathrm{est}} - g_{\mathrm{est}}}{h}.$$

Optimal chattering and the conjectured expansions of Section 5 would give limits $Q/2$ for the exact slopes $(c - g(h))/h$, $(G(h) - c)/h$, and $Q$ for their sum. Finite-word slopes require both search errors to be $o(h)$ to share these limits. Cancellation can conceal those errors, so the table also reports separate slopes at $h = 0.025$. Over the fourteen pairs, the maximum relative deviations of $s_- + s_+$ from $Q$ are 2.52%, 0.64% and 0.31% at $h = 0.1, 0.05, 0.025$, and at the finest step the midpoint $(g_{\mathrm{est}} + G_{\mathrm{est}})/2$ differs from $c$ by at most $0.83h^2$. Theorem 3.12 proves only one-sided bounds; these diagnostics support, but prove neither, matching expansions and optimal chattering, and better words remain possible.

Table 8: Fourteen-pair rate diagnostics (Numerical observation 4.7). Word estimates bound the discrete minima from above up to rounding; slopes use the unrounded repository records.

| | | | $s_- + s_+$ (target $Q$) | | | target $Q/2$ $h = 0.025$ | |
|---|---|---|---|---|---|---|---|
| pair | $a^*$ | $Q$ | $h = 0.1$ | $h = 0.05$ | $h = 0.025$ | $s_-$ | $s_+$ |
| planar | 0.3905 | 1.4780 | 1.4861 | 1.4800 | 1.4785 | 0.7466 | 0.7320 |
| case 1 | 0.7890 | 0.5480 | 0.5619 | 0.5507 | 0.5485 | 0.2830 | 0.2655 |
| $d = 5$, dense 1 | 0.8548 | 0.1785 | 0.1785 | 0.1785 | 0.1785 | 0.0901 | 0.0884 |
| $d = 5$, dense 2 | 0.1906 | 0.5888 | 0.5901 | 0.5898 | 0.5889 | 0.2964 | 0.2926 |
| $d = 5$, dense 3 | 0.2520 | 0.1975 | 0.1973 | 0.1974 | 0.1975 | 0.0989 | 0.0985 |
| $d = 5$, sparse 1 | 0.2068 | 1.3307 | 1.3273 | 1.3282 | 1.3304 | 0.6703 | 0.6601 |
| $d = 5$, sparse 2 | 0.8950 | 0.2024 | 0.2026 | 0.2025 | 0.2025 | 0.1022 | 0.1003 |
| $d = 5$, sparse 3 | 0.4736 | 2.3887 | 2.4183 | 2.4010 | 2.3903 | 1.2025 | 1.1878 |
| $d = 10$, dense 1 | 0.2085 | 0.0447 | 0.0445 | 0.0446 | 0.0447 | 0.0219 | 0.0228 |
| $d = 10$, dense 2 | 0.4063 | 0.3178 | 0.3191 | 0.3183 | 0.3179 | 0.1565 | 0.1615 |
| $d = 10$, dense 3 | 0.4197 | 0.0770 | 0.0768 | 0.0770 | 0.0770 | 0.0386 | 0.0384 |
| $d = 10$, sparse 1 | 0.8253 | 0.7134 | 0.6981 | 0.7091 | 0.7123 | 0.3654 | 0.3469 |
| $d = 10$, sparse 2 | 0.3296 | 0.4656 | 0.4732 | 0.4686 | 0.4671 | 0.2199 | 0.2472 |
| $d = 10$, sparse 3 | 0.5997 | 1.5941 | 1.6197 | 1.6006 | 1.5959 | 0.8187 | 0.7772 |

# D Proof of the rate constants

We keep the notation of Theorem 3.12 and its proof: $m = 2$, $S_a = aA_1 + (1-a)A_2$, the Perron vectors $u_a, v_a$ of $S_a$ with $u_a^\top v_a = 1$, $\kappa(a) = u_a^\top (aA_1^2 + (1-a)A_2^2)v_a$, and the words $w(p,q)$ defined by (3). Every $S_a$ with $0 < a < 1$ is irreducible.

**Lemma D.1.** *Let $\mathcal{I} \subset (0,1)$ be a compact interval. There are $T_1, h_3, C > 0$ such that, for every $h \in (0, h_3]$ and all integers $1 \le p < q$ with $b = p/q \in \mathcal{I}$ and $T := qh \le T_1$, the word $w = w(p,q)$, defined by (3), satisfies*

$$\left|\frac{\log \rho(B_w)}{T} - \eta(b) + \frac{h}{2}\kappa(b)\right| \le Ch^2, \qquad \left|\frac{\log \rho(R_w)}{T} - \eta(b) - \frac{h}{2}\kappa(b)\right| \le Ch^2.$$

*Proof.* Fix $T_1 > 0$ and take $h_3 K_A \le 1/2$. Write $S = S_b$, $\eta = \eta(b)$, $u = u_b$ and $v = v_b$, choosing the Perron vectors continuously with $\mathbf{1}^\top v_a = 1$ and $u_a^\top v_a = 1$. Their norms are bounded on $\mathcal{I}$. We use the induced $\ell^1$ norm; constants $C$ may change from line to line and depend only on $A_1, A_2, \mathcal{I}, T_1$, never on $p, q, h, T$.

Let $\sigma = w_j$ on $[t_{j-1}, t_j)$, where $t_j = jh$, and let $\Phi(t,s)$ be its transition matrix. If $W(s)$ is the time spent in mode 1 up to $s$, put $\delta(s) = W(s) - bs$ and $F = A_1 - A_2$. The prefix counts give $\delta(t_k) = h(\lfloor kb \rfloor - kb)$; since $|\delta'| \le 1$, we have $|\delta| \le 2h$ and $\delta(0) = \delta(T) = 0$. Thus $A_\sigma - S = \Theta'$ almost everywhere, where $\Theta = \delta F = O(h)$.

For $0 \le s \le t \le T$, set $\Theta_s(r) = \Theta(r) - \Theta(s)$. Integrating the identity $\frac{\mathrm{d}}{\mathrm{d}r}(e^{(t-r)S}\Phi(r,s)) = e^{(t-r)S}\Theta_s'(r)\Phi(r,s)$ by parts gives

$$\Phi(t,s) - e^{(t-s)S} = \Theta_s(t)\Phi(t,s) + \int_s^t e^{(t-r)S}\big(S\Theta_s(r) - \Theta_s(r)A_{\sigma(r)}\big)\Phi(r,s)\,\mathrm{d}r.$$

The transition matrices and exponentials have norm at most $e^{K_A T_1}$, while $\Theta_s = O(h)$, so $\|\Phi(t,s) - e^{(t-s)S}\| \le Ch$. Over the full period the boundary term vanishes, giving $\|E_1\| \le ChT$ for $E_1 = \Phi(T,0) - e^{TS}$. Projecting this identity and using $\Phi(r,0)v = e^{r\eta}v + O(h)$ yields

$$u^\top E_1 v = e^{T\eta}\int_0^T u^\top\big(S\Theta - \Theta A_\sigma\big)v\,\mathrm{d}r + O(h^2 T).$$

The replacement error is $O(h^2T)$ because an $O(h)$ vector error multiplies an $O(h)$ factor under the integral. Now $S\Theta - \Theta A_\sigma = \delta[S,F] - \delta\delta' F^2$, and

$$u^\top [S,F] v = \eta u^\top F v - \eta u^\top F v = 0, \qquad \int_0^T \delta\delta' \,\mathrm{d}r = \tfrac{1}{2}\big(\delta(T)^2 - \delta(0)^2\big) = 0.$$

Hence $u^\top E_1 v = O(h^2T)$. Both cancellations hold for every $b \in \mathcal{I}$, not only at a minimizing mixture.

Take $H_j = B_{w_j}$ for every $j$, or $H_j = R_{w_j}$ for every $j$, and set $G_j = e^{hA_{w_j}}$. Taylor and Neumann expansions (minus for explicit Euler and plus for implicit Euler) give

$$D_j := H_j - G_j = \mp h^2 A_{w_j}^2/2 + O(h^3).$$

Since $\|G_j\| \le e^{hK_A}$ and $\|H_j\| \le e^{2hK_A}$, all products over times at most $T_1$ are uniformly bounded. Exploiting the telescopic sum yields

$$E_2 := H_q \cdots H_1 - \Phi(T,0) = \sum_{j=1}^{q} H_q \cdots H_{j+1} D_j G_{j-1} \cdots G_1, \qquad \|E_2\| \le Cqh^2 = ChT.$$

The same estimate on each suffix, combined with the exact-flow comparison, yields

$$u^\top H_q \cdots H_{j+1} = e^{(T-t_j)\eta} u^\top + O(h), \qquad G_{j-1} \cdots G_1 v = e^{t_{j-1}\eta} v + O(h).$$

In the telescoping sum these replacements and the local Taylor remainder contribute $O(h^3)$ per term, hence $O(h^2T)$ in total. As $\sum_j A_{w_j}^2 = q(bA_1^2 + (1-b)A_2^2)$,

$$u^\top E_2 v = \mp e^{(T-h)\eta} \frac{h^2}{2} \sum_{j=1}^{q} u^\top A_{w_j}^2 v + O(h^2T) = \mp e^{T\eta} \frac{hT}{2} \kappa(b) + O(h^2T),$$

where replacing $e^{-h\eta}$ by 1 also costs $O(h^2T)$. Consequently

$$X := H_q \cdots H_1 = e^{TS} + E, \qquad \|E\| \le ChT, \qquad u^\top E v = \mp e^{T\eta} \frac{hT}{2} \kappa(b) + O(h^2T).$$

The eigenvalue gap of $e^{TS}$ shrinks as $T \downarrow 0$. To remove this dependence, compare

$$M_{b,T} := \frac{e^{TS_b} - I}{T}, \qquad \widehat{M}_T := \frac{X - I}{T} = M_{b,T} + \frac{E}{T}.$$

Set $M_{b,0} = S_b$. This is a continuous family of irreducible Metzler matrices on $\mathcal{I} \times [0, T_1]$, with Perron vectors $u_b, v_b$ and eigenvalue $\theta_{b,T} = (e^{T\eta(b)} - 1)/T$, extended by $\theta_{b,0} = \eta(b)$. Its Perron eigenvalue has a uniform real-part gap by irreducibility and compactness. Since $E/T = O(h)$, simple eigenvalue perturbation, after reducing $h_3$, gives uniformly

$$\eta(\widehat{M}_T) = \theta_{b,T} + \frac{u^\top E v}{T} + O(h^2).$$

The Euler factors are nonnegative for $h$, so $X \ge 0$ and $\eta(\widehat{M}_T) = (\rho(X) - 1)/T$. Thus

$$\rho(X) = e^{T\eta} + u^\top E v + O(h^2T) = e^{T\eta}\big(1 \mp \tfrac{hT}{2}\kappa(b) + O(h^2T)\big).$$

The quadratic remainder in the scalar logarithm is $O(h^2T^2)$. Dividing by $T \le T_1$ gives

$$\frac{\log \rho(X)}{T} = \eta \mp \frac{h}{2}\kappa(b) + O(h^2),$$

uniformly even as $T \downarrow 0$, proving both estimates. □

*The frequency average of Remark 3.14.* At the minimizing mixture, set $S = S_{a^*}$, $c = \eta(S)$ and $\Pi = vu^\top$. On the invariant subspace $\ker u^\top$, the matrix $N = (S - cI)|_{\ker u^\top}$ has spectrum in $\operatorname{Re} z < 0$, and for $\omega \neq 0$,

$$(c + \imath\omega - S)^{-1} = \frac{\Pi}{\imath\omega} + (\imath\omega - N)^{-1}(I - \Pi).$$

After multiplication by $u^\top F$ and $Fv$, the first term vanishes because $u^\top Fv = 0$. The scalar identity $\mathrm{PV}\int_{\mathbb{R}}(\imath\omega - y)^{-1}\,\mathrm{d}\omega = \pi$ for $\operatorname{Re} y < 0$ gives $\mathrm{PV}\int_{\mathbb{R}}(\imath\omega - N)^{-1}\,\mathrm{d}\omega = \pi I$ on the stable subspace. Real matrices make $\psi$ even, and the leading resolvent term is purely imaginary, so $\psi(\omega) = O(\omega^{-2})$. Since the stable resolvent is bounded near zero, $\psi$ is integrable and

$$\int_0^\infty \psi(\omega)\,\mathrm{d}\omega = \frac{\pi}{2}u^\top F(I - \Pi)Fv = \frac{\pi}{2}u^\top F^2 v.$$

Multiplying by $2a^*(1 - a^*)/\pi$ gives the formula for $Q$ in Remark 3.14.

*Three or more modes.* For a word of period $T > 0$, let $W_i(s)$ be the occupation time of mode $i$ up to $s$, and set $b_i = W_i(T)/T$ and $S = \sum_i b_i A_i$. Assume that $S$ is irreducible, and let $u, v$ be its left and right Perron vectors, normalized by $u^\top v = 1$. Set $\tilde{F}_i = A_i - A_m$ and $\delta_i(s) = W_i(s) - b_i s$ for $i < m$. Then $A_\sigma - S = \Theta'$ with $\Theta = \sum_{i<m}\delta_i\tilde{F}_i$ and $\Theta(0) = \Theta(T) = 0$. The Perron projection of $[S, \Theta]$ still vanishes, but the remaining term is

$$-\int_0^T u^\top \Theta\Theta' v\,\mathrm{d}r = -\frac{1}{2}\sum_{i<k}\mathcal{A}_{ik}\,u^\top[\tilde{F}_i, \tilde{F}_k]v, \qquad \mathcal{A}_{ik} = \int_0^T (\delta_i\delta_k' - \delta_k\delta_i')\,\mathrm{d}r,$$

where $1 \le i < k < m$. Integration by parts eliminates the diagonal terms and pairs the others into commutators. Here $\mathcal{A}_{ik}$ is twice the signed area of the discrepancy path in coordinates $i, k$. If $\delta_i = O(h)$, boundedness of $\delta_i'$ gives $\mathcal{A}_{ik} = O(hT)$. This term depends on letter order and need not vanish. Thus, for three or more modes, an additional ordering-dependent contribution must be controlled when identifying the first-order coefficient.

# E Proof of the restriction lemma

*Proof of Lemma 2.9.* Fix a nonempty $J$ and $\mu \in \mathcal{M}$. For simplicity, we divide the proof.

*Proof of (a) and (b).* The set $X_J$ need not be closed, so invariance of $\mu$ does not by itself imply invariance of its restriction. The following cutoff estimates show both that no mass flows out of $F_J$ and that the boundary terms vanish when restricting to $X_J$. Set $s(p) = \sum_{k\notin J} p_k$. Then $\nabla s \cdot F_i = \ell_i + r_i$, with the outflow

$$\ell_i(p) = \sum_{k\notin J,\, j\in J}(A_i)_{kj}p_j \ge 0, \qquad |r_i(p)| \le 2K_A s(p).$$

Similarly, for $j \in J$, $\nabla p_j \cdot F_i = \iota_{ij} + r_{ij}$ with

$$\iota_{ij}(p) = \sum_{l\neq j}(A_i)_{jl}p_l \ge 0, \qquad |r_{ij}(p)| \le 2K_A p_j.$$

For $\varepsilon > 0$ consider the $C^1$-regular cut-off function $\psi_\varepsilon(r) = (r/\varepsilon)(2 - r/\varepsilon)$ on $[0, \varepsilon]$, $\psi_\varepsilon = 1$ on $[\varepsilon, \infty)$ and $\psi_\varepsilon(r) = 2r/\varepsilon$ for $r < 0$. It satisfies $0 \le \psi_\varepsilon' \le 2/\varepsilon$ on $[0, \infty)$, $\psi_\varepsilon'(0) = 2/\varepsilon$, and $\psi_\varepsilon'(r)\,r \le 2(r/\varepsilon)\mathbf{1}_{r<\varepsilon}$ for $r \ge 0$ as well as $0 \le \psi_\varepsilon \le 1$ on $[0, \infty)$. By (2) with $f = \psi_\varepsilon \circ s$,

$$\int \psi_\varepsilon'(s)\,\ell\,\mathrm{d}\mu = -\int \psi_\varepsilon'(s)\,r\,\mathrm{d}\mu \le 2K_A\int \psi_\varepsilon'(s)\,s\,\mathrm{d}\mu \le 4K_A\int \frac{s}{\varepsilon}\mathbf{1}_{s<\varepsilon}\,\mathrm{d}\mu.$$

Hence $\int_{\{s=0\}}\ell\,\mathrm{d}\mu \le 2K_A\varepsilon$, so $\ell = 0$ $\mu$-almost everywhere on $\{s = 0\}$, and $\int\psi_\varepsilon'(s)|\nabla s \cdot F|\,\mathrm{d}\mu \le 8K_A\int\frac{s}{\varepsilon}\mathbf{1}_{s<\varepsilon}\,\mathrm{d}\mu \to 0$ as $\varepsilon \to 0$, by dominated convergence. In the same way $\int\psi_\varepsilon'(p_j)|\nabla p_j \cdot F|\,\mathrm{d}\mu \to 0$ for $j \in J$. This gives (a) and (b):

(a) For $\mu$-almost every $(p,i)$ with $p \in X_J$ we have $i \in S(J)$. Otherwise $(A_i)_{kj} > 0$ for some $k \notin J$, $j \in J$, and $\ell_i(p) \geq (A_i)_{kj} p_j > 0$. In particular $\mu(X_J) = 0$ if $S(J) = \emptyset$.

(b) It holds $\int_{X_J} \nabla f \cdot F \,\mathrm{d}\mu = 0$ for every $f \in C^1(\mathbb{R}^d)$. Apply (2) to $f\chi_\varepsilon$ with $\chi_\varepsilon = (1 - \psi_\varepsilon(s)) \prod_{j\in J} \psi_\varepsilon(p_j)$, which tends to the indicator of $X_J$ pointwise on $\Delta$, with $|\chi_\varepsilon| \leq 1$. Then $|\nabla \chi_\varepsilon \cdot F_i| \leq \psi_\varepsilon'(s)|\nabla s \cdot F_i| + \sum_{j\in J} \psi_\varepsilon'(p_j)|\nabla p_j \cdot F_i|$, whose integral tends to 0.

*Proof of (c).* Let $S(J) \neq \emptyset$. We show $\int_{X_J} \gamma \,\mathrm{d}\mu \geq \check{\lambda}_J \,\mu(X_J)$. If $J = \{j\}$, then $X_J = \{e_j\}$ and $\gamma_i(e_j) = (A_i)_{jj} \geq \check{\lambda}_J$ for $i \in S(J)$, so (a) gives the claim. Let $|J| \geq 2$ and $\nu' < \check{\lambda}_J$, and define, for $x \in F_J$, the function

$$v(x) = \inf\big\{ e^{-\nu' t} \mathbf{1}^\top \Phi_\sigma(t) x \;:\; t \geq 0,\ \sigma \text{ with values in } S(J) \big\}.$$

Then $v$ is concave and $v \leq \mathbf{1}^\top x$. Moreover $v(x) \geq \min_{j\in J} x_j > 0$ on $X_J$, because $\mathbf{1}^\top \Phi_\sigma(t) x \geq \min_{j\in J} x_j \, \|\Phi_\sigma(t)|_{F_J}\|_1 \geq \min_{j\in J} x_j \, e^{\check{\lambda}_J t}$ by Proposition 2.2(i). Finally $v(\Phi_\sigma(t)x) \geq e^{\nu' t} v(x)$ for signals with values in $S(J)$. Fix $\theta \in (0,1]$ and let $w = v^\theta$ on $X_J$, a relatively open convex subset of the affine hull of $F_J \cap \Delta$, whose tangent space $T_J = \{y :\ y_k = 0 \ (k \notin J),\ \mathbf{1}^\top y = 0\}$ contains $F_i(p)$ for $p \in X_J$ and $i \in S(J)$. Since $r \mapsto r^\theta$ is concave and nondecreasing, $w$ is concave, hence locally Lipschitz; and $0 < w \leq 1$. On $\{p \in X_J :\ p_j \geq \delta/2 \ \forall j \in J\}$ we have $v \geq \delta/2$, so $w$ is Lipschitz there. By homogeneity and (1), along the projective solution of a mode $i \in S(J)$, which stays in $X_J$, we have

$$w(p(t)) \geq w(p) \exp\left( \theta \int_0^t (\nu' - \gamma_i(p(s)))\,\mathrm{d}s \right).$$

Since $p(t) = p + tF_i(p) + O(t^2)$ and $w$ is locally Lipschitz, at every point $p$ where $w$ is differentiable this gives

$$\nabla w(p) \cdot F_i(p) \geq \theta(\nu' - \gamma_i(p))\, w(p) \qquad \text{for every } i \in S(J), \tag{E1}$$

with $\nabla w$ the gradient in $T_J$. To use restricted invariance, we cut off near the boundary and mollify $w$ to obtain a $C^1$ test function. The preceding estimates will remove the resulting boundary terms. For $\delta > 0$ let $\zeta_\delta \in C^1(\mathbb{R})$ satisfy $\zeta_\delta = 0$ on $(-\infty, \delta]$, $\zeta_\delta = 1$ on $[2\delta, \infty)$, $0 \leq \zeta_\delta \leq 1$ and $0 \leq \zeta_\delta' \leq 2/\delta$, and let $\kappa_\delta(p) = \prod_{j\in J} \zeta_\delta(p_j)$, so $0 \leq \kappa_\delta \leq 1$. Mollify $w$ in the affine hull of $X_J$, with $w_\omega = w * \rho_\omega$. For $\omega < \delta/2$ it is smooth near $\{\kappa_\delta > 0\} \cap X_J$, $0 < w_\omega \leq 1$ there, and $\nabla w_\omega = (\nabla w) * \rho_\omega$. By (E1) and the Lipschitz continuity of $F_i$, $\gamma_i$ and $w$ on $\{p \in X_J \;:\; p_j \geq \delta/2 \ \forall j \in J\}$, there is $c_\delta$, depending also on $\theta$ and $\nu'$, with

$$\nabla w_\omega \cdot F_i \geq \theta(\nu' - \gamma_i)\, w_\omega - c_\delta \omega \qquad \text{on } \{\kappa_\delta > 0\} \cap X_J,\ i \in S(J).$$

Let $G = \kappa_\delta w_\omega$ on the affine hull of $F_J \cap \Delta$, extended by 0 and note that it is $C^1$-regular. Set

$$H(p) := \phi(\mathbf{1}^\top p_J)\, G\big(p_J / \mathbf{1}^\top p_J\big),$$

when $\mathbf{1}^\top p_J \neq 0$, and set $H(p) = 0$ when $\mathbf{1}^\top p_J = 0$. Here $\phi \in C^1(\mathbb{R})$ vanishes near 0 and satisfies $\phi(1) = 1$, so $H \in C^1(\mathbb{R}^d)$, and on $X_J$, for $i \in S(J)$, $\nabla H \cdot F_i = \nabla G \cdot F_i$, because $\mathbf{1}^\top (F_i)_J = 0$ there. By (b) and (a),

$$0 = \int_{X_J} \nabla H \cdot F \,\mathrm{d}\mu \geq \theta \int_{X_J} \kappa_\delta(\nu' - \gamma)\, w_\omega \,\mathrm{d}\mu - c_\delta \omega\, \mu(X_J) - \sum_{j\in J} \int_{X_J} |\zeta_\delta'(p_j)|\, |\nabla p_j \cdot F| \,\mathrm{d}\mu.$$

Since $|\zeta_\delta'| \leq 8\psi_{4\delta}'$ on $[0,\infty)$, the last sum tends to 0 as $\delta \to 0$, by the estimate above. Letting $\omega \to 0$ and then $\delta \to 0$ gives $\int_{X_J} (\gamma - \nu')\, v^\theta \,\mathrm{d}\mu \geq 0$. Letting $\theta \to 0$, with $v^\theta \to 1$ on $X_J$ and dominated convergence, gives $\int_{X_J} \gamma \,\mathrm{d}\mu \geq \nu' \mu(X_J)$. Now let $\nu' \uparrow \check{\lambda}_J$. □

# F Construction and experimental protocols

This appendix records the procedures behind Section 4. The sample generators and detailed scaling and switching results are in Appendix G.

## F.1 Notation and verification

For a Metzler family $\{A_i\}$, $\check{\lambda}$ is its lower Lyapunov exponent, $g(h)$ the explicit Euler lower bound and $G(h)$ the implicit Euler upper bound, with $B_i = I + hA_i$, $R_i = (I - hA_i)^{-1}$, $\mathcal{B}_h = \{B_i\}$ and $\mathcal{R}_h = \{R_i\}$. Write $S_\alpha = \sum_i \alpha_i A_i$ for a convex mixture. Word estimates are numerical upper estimates of the discrete lower spectral radii, not certified approximations of them. A target gap below such an estimate is not a certified two-sided error for $\check{\lambda}$.

Certified bracket endpoints use exact rational checks and independent Python and MATLAB replay. They hold for both exact-decimal and binary64 readings of the input data. The Gripenberg algorithm of Guglielmi and Maiale is discussed in Appendix B. Its adaptive variant supplies the estimates in Tables 5 to 7; lower estimates require a posteriori certification. A failed or unfinished closure is inconclusive. Timings are wall-clock times from single-threaded runs, several executed concurrently on a 16-core Apple-silicon machine under normal load.

## F.2 Certificate construction and verification

**Lower certificate search** To find a certificate at a target level $t$, we close a seed set under the maps $v \mapsto B_i v/t$. Start with coordinate vectors and add $w = B_i v/t$ when the LP $\max\{\mathbf{1}^\top \mu : V\mu \le w,\ \mu \ge 0\} < 1$ detects $w \notin \operatorname{conv} V + K$. Discard redundant vertices. Termination gives the inequalities in Proposition 4.3 in floating point.

Bisect between a certified level and the smaller of $1 + hc$ and the adaptive Gripenberg word bound, with $c = \min_\alpha \eta(S_\alpha)$. The starting level is the smallest column sum of $B_i$, or an estimate whose closure finishes. Probes stopped by time or vertex caps are inconclusive.

**Exact verification** Rationalized LP weights give an exact level $t_*$; a dual vector excludes invariance above $\tau_{\mathrm{up}}$. All 342 lower certificates have $\tau_{\mathrm{up}}/t_* - 1 < 5 \times 10^{-8}$, measuring sharpness for that polytope, not accuracy for $\check{\lambda}$. Checks are independently replayed.

**Upper word certificates** We search finite words using the settings in Section F.8. We certify the best word by Proposition 4.4: rationalize its Perron vector $x_0$ and $r$, and round $(1 + 2^{-40})R_{w_j} x_{j-1}$ upward to dyadic $x_j$. The inverse-free check raises the word rate by less than $10^{-6}$.

**Mixture and schedule certificates** Mixture bounds use Collatz–Wielandt at a rational $\alpha$ near the numerical minimizer. For optimized two-segment schedules with rational durations, choose a diagonal shift $\gamma$ and enclose $e^{tA_i}$ by the Taylor series of $2^{-j}t(A_i + \gamma I) \ge 0$ with a tail bound, square $j$ times and multiply by an upper bound for $e^{-\gamma t}$, rounding upward. Collatz–Wielandt bounds $\rho(\Phi)$, and an artanh series with remainder bounds $\log \rho(\Phi)/T$. To verify a proposed upper growth rate $U$, the independent replay uses uniformization and compares an upper bound for $\rho(\Phi)$ with a lower bound for $e^{UT}$.

## F.3 Test-pair certificate records

Lower certificates have 169 to 3131 vertices and hold for exact-decimal and binary64 data. Case 1440 at $h = 0.025$ reuses the $h = 0.05$ polytope (Proposition 4.3(ii)). Bisections took 8 to 4391s per row; upper certificates use words of length 5 to 141.

For case 1 at $h = 0.1$, the certified $-1.044234$ is close to the separate floating-point word estimate $-1.044232$.

For case 1440 at every reported step and case 148 at $h \le 0.05$, certified upper bounds beat the numerical mixture minima by up to $2.2 \times 10^{-3}$ and $3.1 \times 10^{-3}$, respectively. This comparison does not prove $\check{\lambda} < c$ without a global lower bound for $c$.

### F.4 Experimental setting

Times refer to Python, except for transposed, extended and larger-sample searches in Numerical observation 4.8, which used MATLAB. Runs were single-threaded on a 16-core Apple-silicon machine, several concurrently. The repository supplies the MATLAB package, complete run records and reproduction instructions.

Appendix B gives the Gripenberg implementation, settings and fixed/adaptive comparison.

Every adaptive Gripenberg lower estimate on the test pairs is certified a posteriori; seven of the 54 random-pair estimates remain uncertified within the budget (Appendix B). Adaptive outputs guide the certificate search, but claimed lower bounds require Proposition 4.3.

### F.5 Algorithm (L) protocol

**Upper bound** Among all products of the matrices $e^{\tau A_i}$ of length at most 16, 14 or 12 (for $d \le 3$, $d = 5$ and $d = 10$), take the product $\Pi$ of length $n$ that minimizes $\rho(\Pi)^{1/n}$. Its rate $\ln \rho(\Pi)/(n\tau)$ is an upper bound for $\check{\lambda}$, computed in floating point.

**Polytope** Close the Perron vector of $\Pi$ under the maps $e^{\tau A_i} e^{-\tau(\beta-\nu)}$, where $\beta$ is that rate, and find by bisection the smallest $\nu \ge 0$ for which the closure finishes. We use the closure code of Proposition 4.3 and the same budgets: 300 s per closure and 1800 s per step.

### F.6 Algorithm (L) comparison conventions

Over steps $0.1, 0.05$, Euler gives the larger lower bound on 9/15 pairs and Algorithm (L) on 6/15 (Table 4, excluding reducible case 148). The tangent test improves neither the planar nor case 1 Euler polytope. Algorithm (L)'s numerical upper bound improves on implicit Euler on 14 pairs; case 1440 uses an Euler word of length 50, beyond its exhaustive search. Certified mixtures or schedules improve both.

Closure code, budgets and hardware are shared, with twelve concurrent runs, but our additional long-word search prevents attributing differences solely to discretization. Our naive implementation may differ from that of [12]; only the Euler lower bounds are independently replayed.

### F.7 Work-precision protocol

Figure 2 uses fixed certified upper bounds: schedules for case 1440 and dimension-5 pair 4 of Table 3, a mixture for case 1. At $h = 0.2, 0.1, 0.05, 0.025$, the explicit-word search supplies a Perron orbit for the levels of the switching sample in Numerical observation 4.8; Algorithm (L) uses $\tau = h$ and 1200 s of closures. Times include searches, failures and checks, with eight concurrent Python runs. MATLAB reproduces the twelve Euler bounds; budget-limited Algorithm (L) results vary slightly by machine.

### F.8 Search settings

**Euler word search** Search all words of length at most 16, 14 or 12 (for $d \le 3$, $d = 5$, $d = 10$), two-block words $1^a 2^b$ with $a, b \le \min(4/h, 400)$, and balanced words of frequencies $p/q$ with $q \le \min(4/h, 320)$, then apply single-letter flips, insertions and deletions.

**Scaling closures** Budgets were 300 s per probe and 1800 s per row, plus 20–300 s for exact checks, with twelve concurrent runs.

**Switching closures** Here $g_{\mathrm{up}}$ is the numerical explicit-word estimate and $\varepsilon$ the target tolerance below it. The search minimizes $\rho(B_w)^{1/|w|}$ over two-block words with durations within four steps of the optimized schedule. The best Perron orbit seeds closures at $1 + h(g_{\mathrm{up}} - \varepsilon)$ for $\varepsilon = 10^{-5}, 10^{-3}, 10^{-2}, 10^{-1}$, with respective budgets $600, 600, 300, 300$ s and at most 20000 vertices. Transposed searches on the 23 rows reaching only 0.1 or no level, with extra targets $0.15, 0.2$ and 300 s per closure, improve 11 rows. For dimension-10 rows still at $\varepsilon \geq 10^{-2}$, extended MATLAB searches use both orientations, five two-block Perron orbits, final vertices from the adaptive Gripenberg algorithm on the transpose at $h = 0.1$, and certified $h = 0.1$ vertices at $h = 0.05$. Targets $0.15, 0.1, 0.05, 0.02, 0.01, 10^{-3}, 10^{-5}$ below the current level receive 1800 s per closure; improvements are marked $^{+}$.

# G Scaling and switching samples

This appendix gives the sample construction and detailed results supporting Numerical observation 4.8; search and verification protocols are in Appendix F.

## G.1 Sample construction and success counts

Transferring certificates improves every direct $h = 0.05$ bound for the original six dimension-5 pairs.

The initial sample has three dense and three sparse pairs in each of dimensions $5, 10$. Dense off-diagonal entries are uniform in $[0, 2/(d-1)]$, diagonals in $[-3, 1]$. Sparse entries are nonzero with probability 0.5 ($d = 5$) or 0.3 ($d = 10$), then log-normal with $\sigma = 1.5$ and expected row sum 2.46; diagonals are uniform in $[-4, 1]$. We require irreducible modes and $\alpha^* \in [0.05, 0.95]$ (13 rejections). No improving two-segment schedule was found; optimal chattering is unproved.

The sample was extended in MATLAB to ten dense and ten sparse pairs per dimension, plus three dense and two sparse pairs in dimension 20 (sparse probability 0.15). All bracket endpoints pass replay. Upper mixture bounds use Collatz–Wielandt, except dimension-10 sparse 8, which uses an improving schedule.

At $h = 0.1$, lower bounds lie within $4.4 \times 10^{-3}$ of the numerical word estimates (Tables 9 and 10) for 18/20 dimension-5 pairs, but for only 1/20 dimension-10 pairs (sparse 7). Dimension-20 certificates use 1845–2810 vertices, with widths 0.33–1.6. Mixture bounds improve on implicit Euler for the original twelve pairs (Table 4).

In these experiments, finer steps increase search cost; bracket widths also include the schedule's suboptimality.

Using the sparse generator above with irreducible modes, we kept pairs whose optimized two-segment schedule beats the numerical value of $c$ by $10^{-3}$, i.e. 16/300 draws in dimension 5 and 28/300 in dimension 10. The first ten of each form Table 3, with best periods 0.6–10.9.

Target tolerance $\varepsilon$ means level $g_{\mathrm{up}} - \varepsilon$ below a floating-point word estimate, not a certified two-sided gap. Rounding the lower endpoint down and word value up each adds at most $10^{-6}$. Schedule bounds and polytopes pass exact checks and replay for both data readings; $\check{\lambda} < c$ remains a numerical comparison. The direct, transposed and extended searches, including seed choices, target levels and budgets, are specified in Section F.8. The lower endpoints in Table 3 report separate searches at each step; retaining transferred certificates prevents the best available lower bound from deteriorating as the step decreases (Proposition 4.3(ii)).

The sample was extended to 30 qualifying pairs in each dimension (among 643 and 343 draws), plus five in dimension 20 (among 76, sparse probability 0.15), using the same MATLAB

protocol without vertices from Gripenberg runs on new pairs. Table 11 completes the switching sample.

At $h = 0.1$, target $10^{-3}$ is reached for all 30 dimension-5 pairs and 14/30 dimension-10 pairs, with median bracket widths $1.4 \times 10^{-2}$ and $3.0 \times 10^{-2}$. At $h = 0.05$, only 5/30 dimension-10 pairs reach $10^{-2}$. Dimension 20 closes only at target 0.1: for all five pairs at $h = 0.1$ and for one at $h = 0.05$, with widths 0.11–0.13.

## G.2 Detailed results

Here $^{\top}$ marks transposition, $^{+}$ extended search, and $^{\dagger}$ transfer from a larger step, justified by Proposition 4.3(ii). A dash denotes failure to close within the budget. Upper bounds and widths are rounded upward; lower endpoints downward.

Table 9: Dimension-5 pairs of Numerical observation 4.8. Here $c$ and $g(h) \leq$ are floating-point mixture and adaptive Gripenberg word estimates, rounded up; $^{*}$time limit for the adaptive algorithm. Lower and upper are certified Euler endpoints (Propositions 4.3 and 4.4); width is their difference and $|V|$ the vertex count. $^{\dagger}$The $h = 0.1$ certificate, valid at $h = 0.05$ by Proposition 4.3(ii), improves the direct finer-step certificate in every row.

| pair ($c$) | $h$ | lower | $g(h) \leq$ | upper | width | $\lvert V\rvert$ |
|---|---|---|---|---|---|---|
| dense 1 ($-0.213383$) | 0.1 | $-0.222751$ | $-0.222636$ | $-0.204815$ | $1.8 \times 10^{-2}$ | 1755 |
| | 0.05 | $-0.222751^{\dagger}$ | $-0.217883^{*}$ | $-0.209005$ | $1.4 \times 10^{-2}$ | 1755 |
| dense 2 ($-0.029331$) | 0.1 | $-0.059850$ | $-0.059459$ | $-0.000609$ | $6.0 \times 10^{-2}$ | 1391 |
| | 0.05 | $-0.059850^{\dagger}$ | $-0.044288^{*}$ | $-0.014798$ | $4.6 \times 10^{-2}$ | 1391 |
| dense 3 ($+0.232972$) | 0.1 | 0.222107 | $0.223019^{*}$ | 0.242747 | $2.1 \times 10^{-2}$ | 1146 |
| | 0.05 | $0.222107^{\dagger}$ | $0.228093^{*}$ | 0.237887 | $1.6 \times 10^{-2}$ | 1146 |
| sparse 1 ($-0.034259$) | 0.1 | $-0.107049$ | $-0.102688$ | 0.029982 | $1.4 \times 10^{-1}$ | 1118 |
| | 0.05 | $-0.107049^{\dagger}$ | $-0.067500^{*}$ | $-0.001609$ | $1.1 \times 10^{-1}$ | 1118 |
| sparse 2 ($-0.102660$) | 0.1 | $-0.641285$ | $-0.113168^{*}$ | $-0.092940$ | $5.5 \times 10^{-1}$ | 1802 |
| | 0.05 | $-0.641285^{\dagger}$ | $-0.107746^{*}$ | $-0.097694$ | $5.5 \times 10^{-1}$ | 1802 |
| sparse 3 ($+0.053208$) | 0.1 | $-0.071028$ | $-0.070350$ | 0.170806 | $2.5 \times 10^{-1}$ | 1454 |
| | 0.05 | $-0.071028^{\dagger}$ | $-0.007714^{*}$ | 0.112334 | $1.9 \times 10^{-1}$ | 1454 |

Table 10: Additional pairs of Numerical observation 4.8. Here $c$ is a certified mixture upper bound, or $^{s}$a certified schedule bound below the numerical mixture minimum. Lower and upper are certified Euler endpoints; $^{\dagger}$transferred from $h = 0.1$, improving the direct certificate. $g(h) \leq$ is the floating-point explicit-word estimate, rounded up. Width is $c$ minus the larger lower bound. Upper bounds and widths are rounded up; all endpoints pass both replays.

| | | | $h = 0.1$ | | | $h = 0.05$ | | |
|---|---|---|---|---|---|---|---|---|
| $d$ | pair | $c$ | lower | $g(h) \leq$ | upper | lower | upper | width |
| 5 | dense 4 | 0.224753 | 0.208178 | 0.208195 | 0.240933 | $0.208178^{\dagger}$ | 0.232908 | $1.7 \times 10^{-2}$ |
| 5 | dense 5 | $-0.102452$ | $-0.109222$ | $-0.109221$ | $-0.096018$ | $-0.109222^{\dagger}$ | $-0.099172$ | $6.8 \times 10^{-3}$ |
| 5 | dense 6 | 0.590670 | 0.561428 | 0.561429 | 0.620237 | $0.561428^{\dagger}$ | 0.605415 | $3.0 \times 10^{-2}$ |
| 5 | dense 7 | 0.553355 | 0.536324 | 0.536325 | 0.570945 | 0.543527 | 0.562075 | $9.9 \times 10^{-3}$ |
| 5 | dense 8 | 0.956855 | 0.940609 | 0.940616 | 0.974893 | $0.940609^{\dagger}$ | 0.965630 | $1.7 \times 10^{-2}$ |
| 5 | dense 9 | 0.338856 | 0.290975 | 0.290976 | 0.386052 | 0.307913 | 0.362520 | $3.1 \times 10^{-2}$ |
| 5 | dense 10 | 0.491873 | 0.476472 | 0.476547 | 0.508426 | $0.476472^{\dagger}$ | 0.500015 | $1.6 \times 10^{-2}$ |
| 5 | sparse 4 | $-0.267567$ | $-0.405495$ | $-0.405494$ | $-0.151325$ | $-0.405495^{\dagger}$ | $-0.207181$ | $1.4 \times 10^{-1}$ |
| 5 | sparse 5 | $-0.111087$ | $-0.279163$ | $-0.279162$ | 0.029473 | $-0.193416$ | $-0.037738$ | $8.3 \times 10^{-2}$ |
| 5 | sparse 6 | 0.236626 | 0.225171 | 0.225360 | 0.246479 | $0.225171^{\dagger}$ | 0.241436 | $1.2 \times 10^{-2}$ |
| 5 | sparse 7 | 2.858514 | 2.544951 | 2.544952 | 3.136416 | 2.703311 | 3.006918 | $1.6 \times 10^{-1}$ |
| 5 | sparse 8 | $-0.389270$ | $-0.474814$ | $-0.415082$ | $-0.368156$ | $-0.474814^{\dagger}$ | $-0.378216$ | $8.6 \times 10^{-2}$ |

Table 10: Continued from the previous page.

| | | | $h = 0.1$ | | | $h = 0.05$ | | |
|---|---|---|---|---|---|---|---|---|
| $d$ | pair | $c$ | lower | $g(h) \le$ | upper | lower | upper | width |
| 5 | sparse 9 | 1.346248 | 1.189855 | 1.189856 | 1.452482 | 1.275491 | 1.403754 | $7.1 \times 10^{-2}$ |
| 5 | sparse 10 | −0.475098 | −0.571418 | −0.569407 | −0.388128 | $-0.571418^{\dagger}$ | −0.430868 | $9.7 \times 10^{-2}$ |
| 10 | dense 4 | 0.396993 | 0.067175 | 0.393396 | 0.401010 | $0.067175^{\dagger}$ | 0.398963 | $3.3 \times 10^{-1}$ |
| 10 | dense 5 | 0.592625 | 0.535052 | 0.579588 | 0.606755 | $0.535052^{\dagger}$ | 0.599577 | $5.8 \times 10^{-2}$ |
| 10 | dense 6 | −0.014617 | −0.199153 | −0.059777 | 0.028727 | $-0.199153^{\dagger}$ | 0.007246 | $1.9 \times 10^{-1}$ |
| 10 | dense 7 | 0.769243 | 0.594096 | 0.759850 | 0.779437 | $0.594096^{\dagger}$ | 0.774244 | $1.8 \times 10^{-1}$ |
| 10 | dense 8 | 1.052014 | 0.732335 | 1.045460 | 1.059949 | $0.732335^{\dagger}$ | 1.055807 | $3.2 \times 10^{-1}$ |
| 10 | dense 9 | 0.004987 | −0.154668 | −0.019725 | 0.029610 | $-0.154668^{\dagger}$ | 0.017302 | $1.6 \times 10^{-1}$ |
| 10 | dense 10 | −0.184297 | −0.341127 | −0.193572 | −0.175481 | $-0.341127^{\dagger}$ | −0.179831 | $1.6 \times 10^{-1}$ |
| 10 | sparse 4 | 0.143602 | 0.029023 | 0.116893 | 0.176427 | $0.029023^{\dagger}$ | 0.159333 | $1.2 \times 10^{-1}$ |
| 10 | sparse 5 | −0.583787 | −0.613114 | −0.591249 | −0.575413 | $-0.613114^{\dagger}$ | −0.579692 | $3.0 \times 10^{-2}$ |
| 10 | sparse 6 | −0.179129 | −0.749589 | −0.205900 | −0.152209 | $-0.749589^{\dagger}$ | −0.165618 | $5.8 \times 10^{-1}$ |
| 10 | sparse 7 | 0.084506 | 0.071806 | 0.073599 | 0.097158 | 0.073149 | 0.090545 | $1.2 \times 10^{-2}$ |
| 10 | sparse 8 | $0.973159^{s}$ | 0.755608 | 0.957096 | 0.989161 | $0.755608^{\dagger}$ | 0.981195 | $2.2 \times 10^{-1}$ |
| 10 | sparse 9 | 0.628628 | 0.344828 | 0.569675 | 0.687165 | $0.344828^{\dagger}$ | 0.658158 | $2.9 \times 10^{-1}$ |
| 10 | sparse 10 | 1.002926 | 0.771833 | 0.944048 | 1.059675 | $0.771833^{\dagger}$ | 1.031721 | $2.4 \times 10^{-1}$ |
| 20 | dense 1 | 0.538291 | 0.210786 | 0.514021 | 0.564454 | $0.210786^{\dagger}$ | 0.551118 | $3.3 \times 10^{-1}$ |
| 20 | dense 2 | 0.519703 | −0.142266 | 0.514174 | 0.525795 | $-0.142266^{\dagger}$ | 0.522678 | $6.7 \times 10^{-1}$ |
| 20 | dense 3 | 0.690889 | 0.117393 | 0.685092 | 0.697866 | $0.117393^{\dagger}$ | 0.694230 | $5.8 \times 10^{-1}$ |
| 20 | sparse 1 | 0.557884 | −0.306752 | 0.524683 | 0.591857 | $-0.306752^{\dagger}$ | 0.574671 | $8.7 \times 10^{-1}$ |
| 20 | sparse 2 | 0.928956 | −0.633672 | 0.884728 | 0.979816 | $-0.633672^{\dagger}$ | 0.953409 | 1.6 |

Table 11: Further pairs of the switching sample of Numerical observation 4.8: 11–30 in dimensions 5 and 10, and five in dimension 20. Upper is a certified schedule bound, and $g(h) \ge$ the better certified lower bound from both orientations. Width is upper minus the larger lower endpoint. $^{\top}$Transposed family; $^{+}$extended search; — no closure. MATLAB bracket endpoints hold for both data readings and pass both replays; $c$ and explicit-word upper estimates are floating point.

| | | | | $h = 0.1$ | | $h = 0.05$ | | |
|---|---|---|---|---|---|---|---|---|
| $d$ | pair | $c$ | upper | $g(h) \le$ | $g(h) \ge$ | $g(h) \le$ | $g(h) \ge$ | width |
| 5 | 11 | 0.564144 | 0.562844 | 0.544576 | 0.544565 | 0.554479 | 0.544478 | $1.9 \times 10^{-2}$ |
| 5 | 12 | 0.077634 | 0.073309 | 0.032674 | 0.032663 | 0.054067 | $0.044066^{\top}$ | $3.0 \times 10^{-2}$ |
| 5 | 13 | 0.497816 | 0.483853 | 0.356192 | 0.356181 | 0.421788 | 0.420787 | $6.4 \times 10^{-2}$ |
| 5 | 14 | −0.815606 | −0.825431 | −0.864856 | −0.864867 | −0.843581 | −0.843592 | $1.9 \times 10^{-2}$ |
| 5 | 15 | 0.115795 | 0.049136 | 0.018823 | 0.018812 | 0.034435 | 0.034424 | $1.5 \times 10^{-2}$ |
| 5 | 16 | 0.436569 | 0.399343 | 0.362758 | 0.362747 | 0.381893 | 0.381882 | $1.8 \times 10^{-2}$ |
| 5 | 17 | 0.774866 | 0.748884 | 0.701671 | 0.701660 | 0.726561 | $0.725560^{\top}$ | $2.4 \times 10^{-2}$ |
| 5 | 18 | 0.448169 | 0.366762 | 0.350265 | 0.350254 | 0.358966 | 0.358955 | $7.9 \times 10^{-3}$ |
| 5 | 19 | 1.565797 | 1.533944 | 1.523421 | 1.523410 | 1.528635 | 1.528624 | $5.4 \times 10^{-3}$ |
| 5 | 20 | 0.715940 | 0.565468 | 0.555534 | 0.555523 | 0.560635 | 0.560624 | $4.9 \times 10^{-3}$ |
| 5 | 21 | 0.234460 | 0.176968 | 0.167223 | 0.167212 | 0.172150 | $0.172139^{\top}$ | $4.9 \times 10^{-3}$ |
| 5 | 22 | 0.711039 | 0.709367 | 0.703899 | 0.703888 | 0.706719 | 0.706708 | $2.7 \times 10^{-3}$ |
| 5 | 23 | −0.243249 | −0.250637 | −0.280925 | −0.280936 | −0.264833 | −0.265834 | $1.6 \times 10^{-2}$ |
| 5 | 24 | 0.713176 | 0.685533 | 0.659217 | 0.658216 | 0.672270 | 0.672259 | $1.4 \times 10^{-2}$ |
| 5 | 25 | 1.271640 | 1.226099 | 1.197536 | 1.197525 | 1.212215 | 1.212204 | $1.4 \times 10^{-2}$ |
| 5 | 26 | 0.092427 | 0.088170 | 0.084819 | 0.084808 | 0.086573 | 0.086562 | $1.7 \times 10^{-3}$ |
| 5 | 27 | 1.043939 | 1.034476 | 1.000730 | 1.000719 | 1.018034 | 1.018023 | $1.7 \times 10^{-2}$ |
| 5 | 28 | 0.926204 | 0.917748 | 0.906064 | 0.905063 | 0.912019 | 0.911018 | $6.8 \times 10^{-3}$ |
| 5 | 29 | 0.182585 | 0.133088 | 0.117201 | 0.117190 | 0.125361 | 0.125350 | $7.8 \times 10^{-3}$ |
| 5 | 30 | −0.224996 | −0.332651 | −0.424698 | −0.424709 | −0.375756 | −0.376757 | $4.5 \times 10^{-2}$ |
| 10 | 11 | 0.795549 | 0.784526 | 0.770406 | $0.750405^{+}$ | 0.777396 | — | $3.5 \times 10^{-2}$ |
| 10 | 12 | 0.130308 | 0.097810 | 0.027943 | 0.027932 | 0.063852 | $0.043851^{+}$ | $5.4 \times 10^{-2}$ |
| 10 | 13 | 0.532599 | 0.508683 | 0.499747 | $0.489746^{\top +}$ | 0.504189 | $0.404188^{\top}$ | $1.9 \times 10^{-2}$ |

Table 11: Continued from the previous page.

| | | | | $h = 0.1$ | | $h = 0.05$ | | |
|---|---|---|---|---|---|---|---|---|
| $d$ | pair | $c$ | upper | $g(h) \leq$ | $g(h) \geq$ | $g(h) \leq$ | $g(h) \geq$ | width |
| 10 | 14 | 0.230392 | 0.213210 | 0.200303 | $0.190302^{\top}$ | 0.207386 | $0.157385^{+}$ | $2.3 \times 10^{-2}$ |
| 10 | 15 | 0.572938 | 0.571814 | 0.563308 | $0.562307^{+}$ | 0.567561 | 0.467560 | $9.6 \times 10^{-3}$ |
| 10 | 16 | 0.940565 | 0.938854 | 0.934184 | $0.933183^{\top}$ | 0.936637 | $0.926636^{\top}$ | $5.7 \times 10^{-3}$ |
| 10 | 17 | 0.840727 | 0.834497 | 0.808640 | 0.808629 | 0.822385 | 0.812384 | $2.3 \times 10^{-2}$ |
| 10 | 18 | 0.063995 | 0.062223 | 0.045147 | $-0.054854^{+}$ | 0.053804 | — | $1.2 \times 10^{-1}$ |
| 10 | 19 | 0.559509 | 0.556613 | 0.551435 | 0.550434 | 0.553980 | $0.533979^{\top+}$ | $6.2 \times 10^{-3}$ |
| 10 | 20 | 0.237440 | 0.214496 | 0.176835 | $0.176824^{\top}$ | 0.195603 | $0.185602^{\top+}$ | $2.9 \times 10^{-2}$ |
| 10 | 21 | 0.836483 | 0.829528 | 0.815532 | 0.805531 | 0.822184 | 0.722183 | $2.4 \times 10^{-2}$ |
| 10 | 22 | 0.820069 | 0.811741 | 0.798140 | $0.788139^{\top+}$ | 0.805005 | $0.705004^{\top}$ | $2.4 \times 10^{-2}$ |
| 10 | 23 | 0.802194 | 0.753531 | 0.748609 | 0.748598 | 0.751044 | 0.651043 | $5.0 \times 10^{-3}$ |
| 10 | 24 | 0.979140 | 0.966475 | 0.964792 | $0.914791^{\top+}$ | 0.965613 | $0.765612^{\top}$ | $5.2 \times 10^{-2}$ |
| 10 | 25 | −0.128046 | −0.133086 | −0.159266 | $-0.160267^{\top}$ | −0.145737 | $-0.155738^{\top}$ | $2.3 \times 10^{-2}$ |
| 10 | 26 | 1.427628 | 1.330134 | 1.304476 | $1.294475^{+}$ | 1.317307 | $1.297306^{\top+}$ | $3.3 \times 10^{-2}$ |
| 10 | 27 | 0.545386 | 0.538185 | 0.408598 | 0.407597 | 0.481903 | $0.471902^{\top+}$ | $6.7 \times 10^{-2}$ |
| 10 | 28 | 0.334115 | 0.271371 | 0.259274 | $0.258273^{\top}$ | 0.265374 | $0.165373^{\top}$ | $1.4 \times 10^{-2}$ |
| 10 | 29 | 0.863320 | 0.694745 | 0.675212 | 0.675201 | 0.685172 | 0.585171 | $2.0 \times 10^{-2}$ |
| 10 | 30 | 1.137295 | 1.135605 | 1.112501 | $1.062500^{+}$ | 1.124068 | $0.974067^{\top}$ | $7.4 \times 10^{-2}$ |
| 20 | 1 | 1.173320 | 1.105091 | 1.096882 | 0.996881 | 1.101188 | — | $1.1 \times 10^{-1}$ |
| 20 | 2 | 0.907818 | 0.906359 | 0.891846 | 0.791845 | 0.899688 | — | $1.2 \times 10^{-1}$ |
| 20 | 3 | 0.718666 | 0.635962 | 0.604406 | 0.504405 | 0.620409 | 0.520408 | $1.2 \times 10^{-1}$ |
| 20 | 4 | 0.689138 | 0.670308 | 0.647472 | 0.547471 | 0.658777 | — | $1.3 \times 10^{-1}$ |
| 20 | 5 | 0.708303 | 0.704794 | 0.691886 | 0.591885 | 0.698495 | — | $1.2 \times 10^{-1}$ |

# References


[1] Blanchini F, Colaneri P, Valcher ME (2011) Is stabilization of switched positive linear systems equivalent to the existence of an Hurwitz convex combination of the system matrices? In: Proc. 50th IEEE CDC–ECC, Orlando, pp 1834–1839, https://doi.org/10.1109/CDC.2011.6160372

[2] Blanchini F, Colaneri P, Valcher ME (2013) A stabilizable switched linear system does not necessarily admit a smooth homogeneous Lyapunov function. In: Proc. 52nd IEEE CDC, Florence, pp 5969–5974, https://doi.org/10.1109/CDC.2013.6760831

[3] Calvez V, Gabriel P, Gaubert S (2014) Non-linear eigenvalue problems arising from growth maximization of positive linear dynamical systems. In: Proc. 53rd IEEE CDC, Los Angeles, pp 1600–1607, https://doi.org/10.1109/CDC.2014.7039628

[4] Colonius F, Kliemann W (1996) The Morse spectrum of linear flows on vector bundles. Trans Amer Math Soc 348(11):4355–4388. https://doi.org/10.1090/S0002-9947-96-01524-3

[5] Colonius F, Kliemann W (2000) The Dynamics of Control. Birkhäuser, Boston, https://doi.org/10.1007/978-1-4612-1350-5

[6] Friedland S, Schneider H (1980) The growth of powers of a nonnegative matrix. SIAM J Algebraic Discrete Methods 1(2):185–200. https://doi.org/10.1137/0601022

[7] Gaitsgory V (2004) On a representation of the limit occupational measures set of a control system with applications to singularly perturbed control systems. SIAM J Control Optim 43(1):325–340. https://doi.org/10.1137/S0363012903424186

[8] Gaitsgory V, Rossomakhine S (2006) Linear programming approach to deterministic long run average problems of optimal control. SIAM J Control Optim 44(6):2006–2037. https://doi.org/10.1137/040616802

[9] Guglielmi N, Maiale FP (2026) A Gripenberg-like algorithm for the lower spectral radius. SIAM J Matrix Anal Appl 47(1):77–106. https://doi.org/10.1137/24M1672456

[10] Guglielmi N, Protasov VY (2013) Exact computation of joint spectral characteristics of linear operators. Found Comput Math 13(1):37–97. https://doi.org/10.1007/s10208-012-9121-0

[11] Guglielmi N, Zennaro M (2020) An antinorm theory for sets of matrices: bounds and approximations to the lower spectral radius. Linear Algebra Appl 607:89–117. https://doi.org/10.1016/j.laa.2020.07.037

[12] Guglielmi N, Laglia L, Protasov VY (2017) Polytope Lyapunov functions for stable and for stabilizable LSS. Found Comput Math 17(2):567–623. https://doi.org/10.1007/s10208-015-9301-9

[13] Jenkinson O (2019) Ergodic optimization in dynamical systems. Ergodic Theory Dynam Systems 39:2593–2618. https://doi.org/10.1017/etds.2017.142

[14] Mejstrik T (2022) The finiteness conjecture for $3 \times 3$ binary matrices. Dolomites Res Notes Approx 15(5):24–38. https://doi.org/10.14658/PUPJ-DRNA-2022-5-3

[15] Mejstrik T, Protasov VY (2025) Stability under dwell time constraints: discretization revisited. Nonlinear Anal Hybrid Syst 57:101608. https://doi.org/10.1016/j.nahs.2025.101608

[16] Molchanov AP, Pyatnitskii ES (1986) Lyapunov functions that define necessary and sufficient conditions for absolute stability of nonlinear nonstationary control systems, I–III. Autom Remote Control 47(3):344–354. Parts II and III: no. 4, 443–451; no. 5, 620–630

[17] Opoitsev VI (1977) Equilibrium and stability in models of collective behaviour. Nauka, Moscow, in Russian

[18] Protasov VY, Jungers RM (2016) Analysing the stability of linear systems via exponential Chebyshev polynomials. IEEE Trans Automat Control 61(3):795–798. https://doi.org/10.1109/TAC.2015.2449151

[19] Rossi F, Colaneri P, Shorten R (2011) Padé discretization for linear systems with polyhedral Lyapunov functions. IEEE Trans Automat Control 56(11):2717–2722. https://doi.org/10.1109/TAC.2011.2161028

[20] Sheng Q (1989) Solving linear partial differential equations by exponential splitting. IMA J Numer Anal 9(2):199–212. https://doi.org/10.1093/imanum/9.2.199

[21] Suzuki M (1991) General theory of fractal path integrals with applications to many-body theories and statistical physics. J Math Phys 32(2):400–407. https://doi.org/10.1063/1.529425

[22] Trotter HF (1959) On the product of semi-groups of operators. Proc Amer Math Soc 10(4):545–551. https://doi.org/10.1090/S0002-9939-1959-0108732-6

[23] Vinter R (1993) Convex duality and nonlinear optimal control. SIAM J Control Optim 31(2):518–538. https://doi.org/10.1137/0331024